Phenotype Strands After SIMB

Expression, Morphology, and Metabolite Production: What Was Measured and What to Run Next

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August 27, 2026

Abstract

Six phenotype strands were run against one shared cell embedding: knockout expression, morphology, the two jointly, betaxanthin production, free amino-acid pools, and beta-carotene production. Two of them were presented at SIMB 2026. This is the record of what each measured, against what ceiling, and with what scoring rule, read from all 28 W&B projects and 2,187 runs rather than from a selection. Most strands sit further from their ceilings than the manuscript's placeholder text assumes. Three findings drive the next campaign. The expression objective is fighting itself: the quantile loss reaches its minimum at epoch 463 and rises for the next 9,500 while Pearson climbs, squared error bottoms alongside the Pearson peak instead, and the model never gets normalized squared error below "predict each gene's mean" because its predictions are twice as spread out as its own correlation justifies, which a free post-hoc rescale fixes. Morphology has never been trained on more than a quarter of its screen, in any of 397 runs across eight projects, and its ceiling is measured here for the first time at 0.611. And the free amino-acid profile of a deletion predicts that deletion's betaxanthin at an out-of-fold r of 0.298 while tyrosine, the chromophore precursor, predicts it at 0.064, so the coupling is real and is not the precursor axis. The closing sections rank what to run next and size the expression-and-morphology campaign in GPU-days.

How to read this. section 1.1 defines the words that recur. section 1 is the whole argument and every number a decision turns on. sections 2 to 6 are one section per strand, each ending in what it established and what it cannot say. section 7 collects the failures the strands share, which is where the leverage is. section 8 ranks the options, and section 9 sizes the next campaign. **Going straight to the campaign:** section 2.3, table 9 and section 9.

Revision 2 answers twelve review comments on revision 1. The ledger is `notes-tex/019-simb-multimodal/review/round-1-dispositions.md`. Two claims in revision 1 were wrong and are corrected in place, both in section 2.3.

Working notes: `notes/experiments.019-simb-multimodal.*.md`, `notes/experiments.020-cachera-betaxanthin.*.md`

Generating scripts: the `scripts/` directory of `experiments/019-simb-multimodal/`, `experiments/020-cachera-betaxanthin/`, and `experiments/023-metabolome-betaxanthin-joint/`

Leaderboard: `experiments/019-simb-multimodal/results/round_leaderboards.csv`

Section status: ✓ ready ■ author-reviewed, keep stable ✗ not done, agents may edit freely

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1 Summary ✗

1.1 Terms used throughout ✗

Head The output module for one phenotype. The encoder and the perturbation operator are shared; a head is the last few layers that turn the shared representation into one phenotype's values. A run declares which heads are active, so *betaxanthin head* and *morphology head* name output modules, not models.

Metabolome head The head that predicts the 19 Mulleder amino-acid concentrations. It appears in two roles. As the **only** active head it is the amino-acid strand (section 5). Added **alongside** the betaxanthin head it is an **auxiliary head**: it is trained and scored, but the number being reported is still betaxanthin, and the question is whether sharing an encoder with it helps.

Auxiliary head Any head added to a run for its effect on a different head's score. Adding one changes two things at once, the gradient signal and the population of strains the run can use, which is why every comparison here pins the instance set.

Strand One phenotype prosecuted across a series of W&B projects.

`pearson_per_feature` For each output feature (a reporter gene, a CalMorph parameter, one amino acid), the correlation across strains between prediction and measurement; then averaged over features. It goes to zero when a model predicts each feature's mean. The companion `pearson_per_instance` correlates across features within one strain and stays high under exactly that collapse, so it is never the objective here.

`nmse` Mean squared error divided by the variance of the target about each feature's mean, so `nmse` = 1 is exactly "predict each gene's training mean".

`roll_max` The scoring rule: the maximum of a centered five-point rolling mean of a validation curve. An upward-biased order statistic whose bias grows with the number of epochs run, so every score here travels with its epoch count.

Ceiling The highest correlation any predictor could reach against a target carrying measurement noise, estimated per strand from replicate structure.

1.2 Where each strand stands ✗

strand	ceiling	best	of ceiling	epochs (peak/last)	runs
Expression (Kemmeren, Sameith)	0.775	**0.229**	30%	4007/4105	857
Expression, masked-label objective	0.775	**0.241**	31%	9188/9997	16
Morphology (CalMorph, Ohya)	0.611	**0.082**	13%	27/65	183 (1 flat)
Expression and morphology, joint	–	**0.059**	–	16/27	214
Betaxanthin (Cachera)	0.914	**0.401**	44%	74/124	142
Betaxanthin with metabolome head	0.914	**0.430**	47%	176/496	46
Amino-acid pools (Mulleder)	–	**0.211**	–	402/999	97
Beta-carotene (Ozaydin)	0.544	**0.132**	24%	799/999	136

Table 1: Every strand, scored the same way. *best* is `roll_max`, the maximum of a centered five-point rolling mean of the validation curve, an upward-biased order statistic whose bias grows with epochs run, which is why the epoch column travels with it. *epochs (peak/last)* gives where the best run peaked and where it stopped: a peak in the last tenth means the run was cut off while still improving. Ceilings are measured by three different estimators because the three measurement types require them; the amino-acid cell is empty because Mulleder has one replicate per strain, and no ceiling is estimable. *flat* counts runs whose validation metric never left zero. **Coverage: 27 of 28 projects, 1691 runs.** The rest are queued (table 19); a strand maximum can only move upward when they land, so every *of ceiling* figure is a lower bound. The betaxanthin row is known to be one such case: table 12 carries 0.4301 from a study in a queued project.

Read the epoch column with the score column. Expression and beta-carotene peak in the last tenth of their best run, so those numbers are lower bounds set by the compute budget. Morphology and the expression-morphology joint arm peak at epochs 27 and 28 of runs that stopped at 65 and 70, which is not a result of any kind. Betaxanthin and amino acid peak in the first half of a 999-epoch run and then overfit.

The two betaxanthin rows are not comparable to each other: the joint row trains on 3,832 strains against the plain row's 3,698, because the joint configuration restricts to genotypes carrying both phenotypes. The within-grid paired comparison in table 16 is the valid form of that contrast, and it comes out negative.

1.3 Eight findings ✗

1. **The expression strand contains no valid comparison between mechanisms.** All 67 scored arms stopped at or below 300 epochs, against a validation curve that dips at 200 to 300 and then climbs past its early peak. No peak has been observed at any budget up to 10,000 epochs. Every per-arm delta on record measures early-training transients.

2. **The expression objective is fighting itself, and it is the quantile loss rather than squared error.** On the longest run the quantile loss bottoms at epoch 463 and rises for the next 9,500 while Pearson climbs, but squared error bottoms at epoch 9,175, alongside the Pearson peak at 9,674; on the second-longest run the two are one epoch apart. `nmse` never returns below 1 after about epoch 400, so the model reaches $r = 0.236$ while being *no better than predicting each gene's mean* in squared error. The cause is calibration, not capability: at the peak its predictions are 1.95× more spread out than its own correlation justifies, and a post-hoc rescale by r/s moves `nmse` from 1.010 to 0.944 while changing no correlation at all. Best-by-`mse` checkpointing would have picked within 500 epochs of the right model, so checkpointing is a narrower problem than previously recorded and is not what holds the strand at 0.24.

3. **One expression run is a multi-day object, and this sets the campaign.** The best run took **91.4 hours, 3.81 days**, for 9,999 epochs at 32.9 s/epoch, and still peaked at 97 % of the way through. One GPU delivers $\approx$ 2,500 epochs per day, so the number of arms a campaign can afford is fixed before any of them are chosen.

4. **The perturbation operator provably cannot express a pair term, and masked unmasking does not supply one.** At one perturbed gene the cross-attention softmax runs over a one-element key set, so the model reduces to $g(h_i + c_b)$ and nothing depends on which reporter is predicted given which gene was deleted. 16,200 of 32,760 attention parameters are dead. The scGPT-style scheme conditions on *other genes' measured values*, and at $k = 0$, which is where scoring happens, every one of those features is exactly zero and the forward pass is identical to the unconditioned model. The two are different axes. Capacity is not the binding constraint either: a rank-32 reconstruction of the target reaches 0.727. A rank-64 pair term already reaches 0.2274 in 2.0 days against the masked objective's 0.2362 in 3.8.

5. **Morphology has never been trained on its own data.** The Ohya screen holds 4,718 deletions; every morphology run used the 1,440 that also carry expression, giving $n_{\text{train}} = 1{,}161$. Its ceiling, measured here for the first time, is **0.611** over 278 features, with 201 of them individually above 0.5. For a second-label experiment, fitness shares 4,220 strains with morphology against expression's 1,440.

6. **The free amino-acid profile predicts betaxanthin; the precursor does not.** Over the 4,432 deletions the two screens share, the 19-dimensional pool predicts betaxanthin at out-of-fold $r = \mathbf{0.298}$, while tyrosine alone, the chromophore precursor, predicts it at **0.064** and correlates marginally at −0.076. Roughly three quarters of the profile's power survives regressing out single-mutant fitness. On the 724 deletions with no Costanzo fitness record, which carry twice the betaxanthin spread, the profile predicts at 0.455.

7. **The metabolome auxiliary head currently costs betaxanthin performance.** Paired within grid cell, joint minus control is -0.0265 ± 0.0159 over the five comparable cells where the control learned the task, with 2 of 5 positive. The +0.0203 that motivated the replication reproduces, at +0.0192, in the one cell it came from. So the coupling exists in the data and the current head does not use it.

8. **Betaxanthin is the strongest case, and its strength is coverage.** yeast-GEM reaches 19 % of the screen and misses 73 % of its high producers. On genes a flux-based method has no features for, CGT's top 25 is enriched 4.4× over base rate. Where both methods can be scored they are tied, they find largely different genes (9 shared of 29 each), and their union reaches 49 of 100.

9. **Configuration choice moves results more than method choice.** Betaxanthin grid cells span Spearman 0.013 to 0.158 against a between-method gap of 0.0014 on AUC. The amino-acid strand's best run is 0.211 and its median is 0.0498.

1.4 The decision ✗

- **Figure 6 is ready to be argued** on coverage rather than on accuracy, and the amino-acid material belongs in it as a coupling measurement, not as a joint-training result.

- **Figure 3 is not ready, and its placeholder numbers are far from the measured ones.** A per-gene Pearson of 0.24 against a ceiling of 0.775 is not a headline. The cheapest path to a better one is not another architecture ladder: it is training morphology on its full dataset, finding where the expression curve actually turns, and checking whether the graph prior is the right prior at all.

- **Three cheap experiments are unrun and each would change what to do next**: the graph-prior probe, morphology at full scale, and one expression run trained until the validation curve turns.

2 Expression ✗

2.1 The task and its ceiling ✗

The **expression strand** predicts, for a strain carrying one gene deletion, the $\log_2$ ratio of every measured gene's transcript abundance against wild type. Targets come from the Kemmeren 2014 knockout microarray compendium (doi:10.1016/j.cell.2014.02.054) and the Sameith 2015 double-deletion compendium (doi:10.1038/sdata.2015.15), fused in the `fig3_core` build. Of the 1,556 records in that build, 1,484 are single deletions and 72 are Sameith doubles, so 95.4 % of the training signal carries exactly one perturbed gene. SRC: `experiments/019-simb-multimodal/results/fig3_overlap_census.json`

The **reproducibility ceiling** is the highest correlation any predictor could reach against a target that carries measurement noise. It is estimated here from the 82 deletions measured independently in both compendia, as the mean over reporter genes of $\sqrt{\mathrm{clip}(\hat{\rho}_g, 0, 1)}$ where $\hat{\rho}_g$ is that gene's test-retest correlation across the paired strains. The estimate is **0.7746**. SRC: `experiments/019-simb-multimodal/results/expression_ceiling_replicate.json`

The best validation score observed on this task is `pearson_per_feature` = **0.2407**, from run `hx8pxdic` in project `torchcell_019_expr_v9`, whose peak sits at epoch 9,188 of the 9,997 it reached. The best in `torchcell_019_expr_v8` is 0.2293, run `b50f93ju`, peaking at epoch 4,007 of 4,105. Both are $\approx$ 30 % of the ceiling, and both peak in the last tenth of their run. SRC: `experiments/019-simb-multimodal/results/round_leaderboards.csv`

The metric is the per-reporter correlation across strains, averaged over reporters, which is the quantity that goes to zero when a model predicts each gene's mean; the companion per-strain metric stays near 0.4 under exactly that collapse and is not the objective. The scoring rule is `roll_max`, the maximum of a centered five-point rolling mean of the validation curve. It is an upward-biased order statistic whose bias grows with the number of epochs run, so it is comparable only between runs of similar length, which is why the epoch count travels with every number quoted from the leaderboard.

The manuscript's Results section currently states $r = 0.543$ for expression. That string sits inside a paragraph marked `[FILLER - R3.]` and is a placeholder, not a measurement.

2.2 The finding of the round: nothing was trained to saturation ✗

Sixty-seven runs across waves 1 to 4b were scored, and not one of them ran past 300 epochs. Against a validation curve that dips at epochs 200 to 300 and then climbs past its early peak, every one of those runs was stopped inside the dip. SRC: `experiments/019-simb-multimodal/results/decoder_arms_torchcell_019_expr_v8.csv`

wave	runs	epoch range
wave 1	10	70–140
wave 2	5	76–140
wave 3	36	18–276
wave 4a	8	77–81
wave 4b	8	300
all	**67**	**18–300, none above 300**

Table 2: Every scored arm in the expression round stopped at or below 300 epochs.

Four long runs were later allowed to continue. All four died at 12.60 to 12.61 hours of wall clock at 1,620 to 1,625 epochs, which is a scheduler limit rather than convergence, and their validation peaks sit at 93 to 99 % of the way through the run. Eval-mode training Pearson reached 0.652 to 0.725 and was still climbing when they were killed.

Longer runs have since been allowed. The picture did not change: the best v8 run peaks at epoch 4,007 of 4,105 and the best v9 run at epoch 9,188 of 9,997. Raising the epoch budget from 1,600 to 10,000 moved the score from 0.204 to 0.241 and moved the peak with it, so the curve is still rising where it was cut off. **No peak has ever been observed on this task**, at any budget tried, and the saturation epoch remains unknown.

Read across all nine expression rounds, the score is a function of the budget. Every expression project, read the same way and sorted by score, comes out in almost exactly the order of its epoch budget.

project	runs	best	epochs of that run	n_{train}
`expr_v9`	16	**0.2407**	9,997	1,244
`expr_v8`	163	**0.2293**	4,105	1,244
`expr_v7`	295	**0.1631**	152	1,244
`expr_v6`	158	**0.1628**	124	1,244
`expr_v2`	120	**0.0996**	25	1,161
`expr_v3`	60	**0.0874**	20	1,161
`expr`	26	**0.0822**	19	1,126
`expr_v5`	35	**0.0719**	27	1,161

Table 3: Every expression round, read the same way and sorted by score. The order is almost exactly the order of the epoch budgets, across rounds that were varying capacity, targets, graphs, decoders and objectives. Whatever those rounds were testing, the thing that moved the number was how long the run was allowed to go. Training-set size moved by only ten percent across the same span and does not order the column.

	v9 `M_fine` (masked)		v8 `V_basis64`	
quantity	value	epoch	value	epoch
val `loss` minimum (quantile)	0.2376	**463**	0.2610	**141**
val `mse` minimum	0.0391	**9,175**	0.0397	**3,922**
val Pearson maximum	0.2362	**9,674**	0.2274	**3,921**
val `nmse` minimum	0.9981	213	0.9925	145
val `nmse` at the Pearson peak	1.0100	–	1.0395	–
run length (epochs)	9,999		4,106	
wall clock	**91.4 h (3.81 days)**		47.9 h (2.00 days)	
seconds per epoch	32.9		42.0	

Table 4: The quantile loss bottoms in the first few hundred epochs and rises for the rest of the run. Squared error bottoms alongside the Pearson peak, within 500 epochs on v9 and within one epoch on v8. Both runs replicate the pattern, and they use different mechanisms and different epoch budgets. **Why the Pearson values differ slightly from table 20**: this table smooths over 25 logged points to *locate* turning points, the leaderboard over 5 to *score* them, so v9 reads 0.2362 at epoch 9,674 here and 0.2407 at 9,188 there. The leaderboard value is the one quoted as a score; these are locators.

These rounds were not varying the same thing. They varied capacity, target normalization, which compendium was included, graph regularization, decoder family, distributional head and objective. None of that orders the column; the epoch budget does, across a factor of 500 in budget and a factor of 3.3 in score. Training-set size moved by ten percent over the same span and does not order it either.

This is the strongest available evidence that the binding constraint on this strand has been compute rather than mechanism, and it is only visible with every round in one table. It is also a caution about reading it too hard: `roll_max` is an upward-biased order statistic whose bias grows with epochs run, so part of the ordering is the scoring rule rewarding longer runs. That bias is a few thousandths at these lengths and the spread here is 0.07 to 0.24, so it explains a sliver rather than the trend.

Consequence. An arm comparison at 300 epochs is not a small effect measured noisily. Both arms were stopped before either had the chance to separate, so the difference between them reflects early-training transients. The null sink arm reads $+0.0024 \pm 0.0090$ over four seed pairs and post-perturbation mixing reads -0.0023 ± 0.0076, and neither number is evidence about its mechanism. More seeds do not repair this: replicates buy precision on the wrong estimand.

2.3 The quantile loss turns early; squared error does not ✗

Correction. An earlier version of this section said loss and metric "move in opposite directions". That is true of the *quantile loss* and false of squared error, and collapsing the two hid the finding. The two longest runs ever trained were pulled and read directly. SRC: `experiments/019-simb-multimodal/scripts/expression\_objective\_diagnosis.py`

The sharper finding: the model never beats the mean in squared error. `nmse` is normalized so that 1 is exactly "predict each gene's training mean". It dips barely below 1 at epoch 213 and is back above 1 from about epoch 400 onward, reading 1.010 on v9 and 1.040 on v8 at their Pearson peaks. So the model reaches $r = 0.236$ while being no better than the mean predictor on a squared-error scoreboard.

That is arithmetic rather than a mystery. Writing s for the ratio of prediction standard deviation to target standard

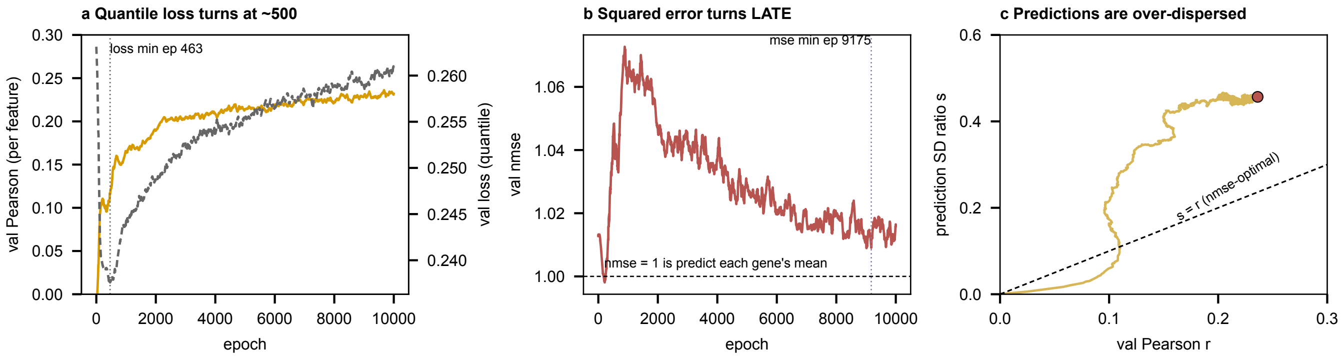


Figure 1: The v9 run, 10,000 epochs. (a) The quantile loss reaches its minimum at epoch 463 and rises thereafter while Pearson climbs. (b) `nmse` dips just under 1 at epoch 213, rises to 1.073 near epoch 900, then falls back for the rest of the run without ever returning below 1; squared error's own minimum is at epoch 9,175. (c) The prediction standard-deviation ratio s against the correlation r, with the `nmse`-optimal line $s = r$.

deviation and r for the correlation, predictions that are a scaled version of a correlated signal give

$$\texttt{nmse} = 1 + s^2 - 2rs,$$

which is minimized at $s^\star = r$ with value $1 - r^2$. The identity reproduces the logged `nmse` to within 0.016 on both runs. At its Pearson peak v9 sits at $s = 0.460$ against $r = 0.236$, so its predictions are **1.95**× more spread out than its own correlation justifies; v8 is 2.21×.

Consequence, and it is free. Multiplying predictions by r/s changes no correlation at all and moves `nmse` from 1.010 to $1 - r^2 = 0.944$ on v9, and from 1.040 to 0.949 on v8. The ordering the model produces is already worth more than its magnitudes, and any report that quotes Pearson beside an `nmse` above 1 is quoting a model that would lose to the mean until it is rescaled.

What this does and does not say about checkpointing. Best-by-loss on the *quantile* loss picks a model ≈ 9,200 epochs early, which is the defect that motivated best-by-metric checkpointing. Best-by-`mse` would have picked within 500 epochs of the right one on v9 and within one epoch on v8. So checkpointing is a narrower problem than it looked, and it is not what is holding the strand at 0.24.

Still not measured. Whether training on a different objective, rather than rescaling after the fact, reaches a higher Pearson. section 8 sets that as an arm rather than an assumption.

2.4 The perturbation operator has no pair term ✗

At $|S_b| = 1$ the cross-attention that carries the perturbation runs its softmax over a one-element key set, so the attention weight is identically 1. Re-drawing W_Q and W_K at standard deviation 10 changes the output by exactly 0.0, and 16,200 of 32,760 attention parameters are dead. SRC: `experiments/019-simb-multimodal/results/perturbation_selector_degeneracy.json`

The model therefore reduces to $g(h_i + c_b)$: gene identity i and strain identity b meet once, by addition, and no term depends on the pair (p, i) of deleted gene and reporter gene. The rank of the pair term each candidate mechanism can express is what wave 6 was built to ladder.

mechanism	form	pair rank
`V_ref` additive	$g(h_i + c_b)$	**0**
`V_sink` gated attention	one bounded scalar per head	9
`V_basis`$\{16, 32, 64\}$	$\sum_j b_{ij}(h_i)\, a_j(z_S)$	r
`V_film`, `V_hadamard`	diagonal multiplicative	$d = 90$

Table 5: Pair-rank ladder. The reference arm cannot express any dependence on which reporter is being predicted given which gene was deleted.

Capacity is not what binds. A rank-r reconstruction of the target matrix reaches 0.7265 at $r = 32$ and 0.7799 at $r = 64$, against a best model score of 0.241, so a null on this axis is a statement about the mechanism. SRC: `experiments/019-simb-multimodal/results/lowrank_output_ceiling.json`

Masked unmasking does not supply the pair term, and this is provable rather than arguable. The scGPT-style scheme conditions gene i's token on the measured values of other genes in the same strain:

$$h_{b,i}^{\text{obs}} = h_{b,i}^{\text{pert}} + g_{\text{obs}} \cdot \text{MLP}\big(\big[\, y_{b,i}\, m_{b,i},\ m_{b,i} \,\big]\big),$$

where $m_{b,i}$ marks gene i as revealed. At $k = 0$ the revealed set is empty, every encoded feature is exactly zero, and the forward pass is **identical** to the unconditioned model. Scoring happens at $k = 0$. So the added term contributes nothing at the point the number is taken, and it cannot create a dependence on the pair (deleted gene, reporter gene) no matter how well the unmasking is trained.

The two are different axes and the round conflated them:

- **Masked conditioning** answers *given some of this strain's genes, what are the rest.* Its channel is other genes' measured values, and table 6 shows that channel is worth a great deal and is measured to be orthogonal to genotype.
- **The pair term** answers *given which gene was deleted, which reporters move.* Its channel is the deletion identity, which is present at $k = 0$.

What can carry a pair term at $k = 0$ is the table 5 ladder: a multiplicative or FiLM conditioner ($h_i \odot (1 + \gamma(z_{S_b}))$, rank $d = 90$), a factored readout ($\sum_j b_{ij}(h_i)\, a_j(z_{S_b})$, rank r), or masked graph attention applied **after** the perturbation operator so that gene i learns about the deletion through its own neighbors. The last of these is the only one that makes the pair term a function of network distance, and it is the one that has never been built.

Read the two best runs together, carefully. The best masked run reaches 0.2362 at epoch 9,674 of 9,999; the best pair-rank-64 run reaches 0.2274 at epoch 3,921 of 4,106. That is a 0.009 gap at a 2.4× difference in epoch budget, so it is not an arm comparison. What it does establish is that a rank-64 pair term gets within 4 % of the masked objective for 40 % of the compute, which is the cheapest existing evidence that the pair axis is worth the next round.

2.5 The largest signal in the strand is imputation, and it is orthogonal ✗

A ridge oracle that observes m randomly chosen genes of a held-out strain and predicts the rest reaches far past anything the model does.

m observed	within-Kemmeren	within-Sameith	Kem → Sam	Sam → Kem
10	0.4562	0.3649	0.2335	0.2122
100	0.6693	0.6417	0.3815	0.3922
1000	0.7832	0.7779	**0.4838**	**0.4803**

Table 6: Masked-conditioning oracle, `pearson_per_feature` on validation strains. The within-study value at $m = 1000$ exceeds the 0.775 reproducibility ceiling, which the cross-study column resolves: same-array conditioning was predicting the target's own measurement noise.

The gain is not a better genotype-to-expression map. Removing a genotype predictor, a k-nearest-neighbor model on `prot_T5_all` with $k = 25$, leaves 97.5 to 100.6 % of the conditioning gain intact, so the channel the objective optimizes is switched off at $m = 0$, which is where the strand is scored. SRC: `experiments/019-simb-multimodal/results/conditioning_gain_after_genotype.json`

What makes the oracle worth keeping is the structure it exploits. After removing each gene's mean, the residual gene-gene correlation pattern replicates on held-out strains at split-half $r = 0.8687$ against a permutation null of 8.45×10^{-5}, and it has an effective rank of 32.78, with 59.1 % of variance inside the top 32 components. A per-gene independent readout emits 6,127 marginals and discards all of it. SRC: `experiments/019-simb-multimodal/results/residual_covariance_diagnostic.json`

2.6 Why the byte count is misleading ✗

The expression compendium holds more scalar labels than the trigenic interaction data the model does well on, which is the intuition that predicted this task would be easy. The label count is not the axis that matters.

In the trigenic data a given gene appears in many records paired with different partners, so the supervision repeatedly constrains that gene's contribution. In the expression data a gene appears as the perturbation exactly once, and it appears in one array. The 6,169 labels attached to it are 6,169 readouts of a single perturbation event, correlated

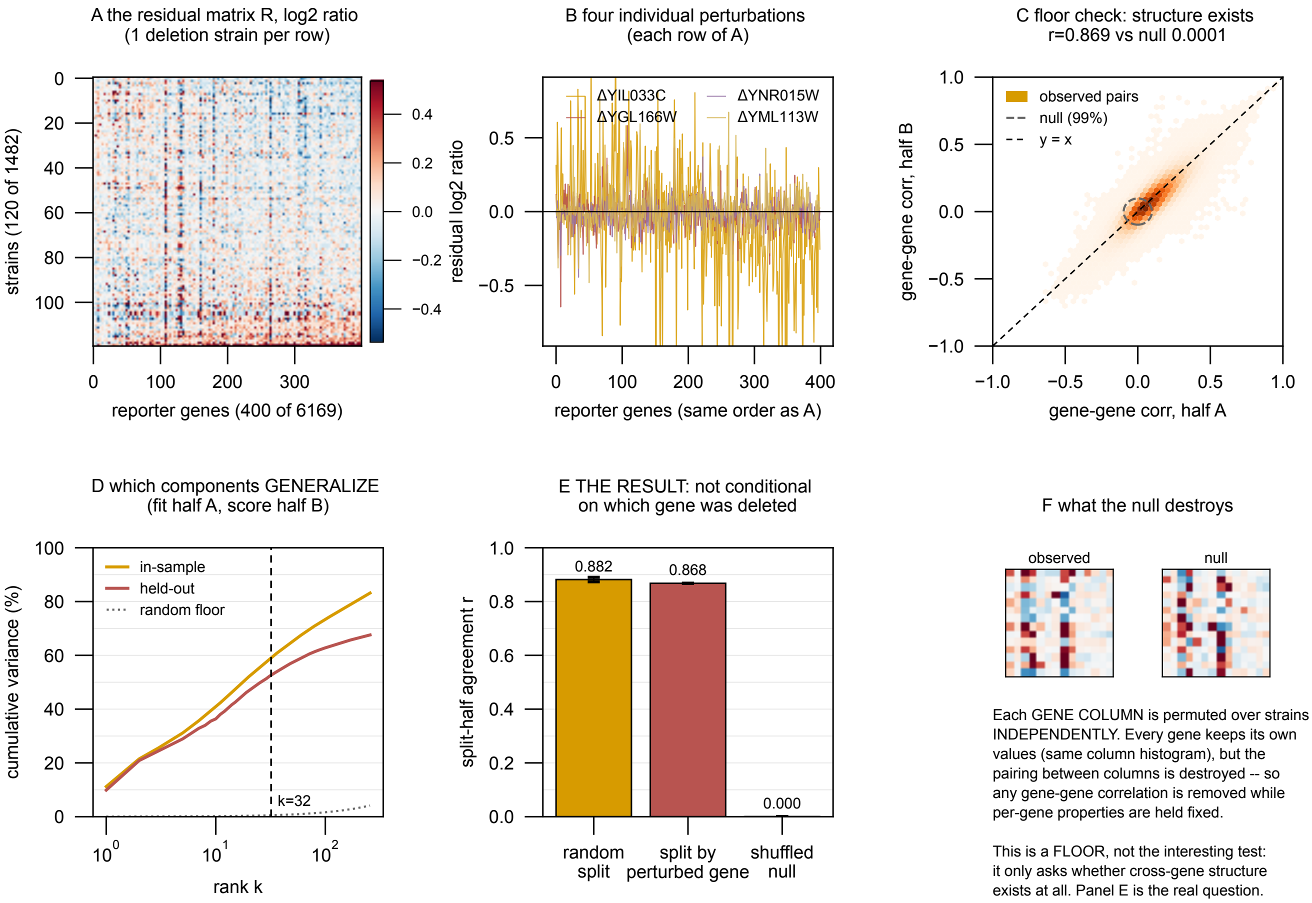


Figure 2: Residual gene-gene structure in the expression compendium after removing each gene's mean. The pattern replicates on held-out strains and lives in about 33 effective components, which is what a per-gene independent readout discards.

dataset	records	per record	scalars	perturbation axis
Kuzmin 2018 trigenic interaction	91,111	1	91,111	
Kuzmin 2020 trigenic interaction	301,798	1	301,798	
both, as 010 queries them	**392,909**	**1**	**392,909**	triples; a gene recurs with many partners
Kemmeren 2014 knockout expression	1,484	6,169	9,153,196	1,484 single deletions, each once
Ohya 2005 morphology	4,718	281	1,325,758	4,718 single deletions, each once

Table 7: Scalar labels are not the binding resource, and the trigenic comparison is with **both** Kuzmin papers: `experiments/010-kuzmin-tmi/queries/001_small_build.cql` selects `TmiKuzmin2018Dataset` and `TmiKuzmin2020Dataset`, 392,909 records between them. Expression still carries more scalars, 9.2 million against 0.39 million, but its perturbation axis has 1,484 points and no gene is deleted twice, so nothing in the data isolates the effect of a deletion from the effect of the one array it was measured on. Record counts are raw rows in each dataset's `preprocess/data.csv`, before the deduplication and aggregation the graph build applies.

with each other at the level the residual covariance above measures, so they are worth far less than 6,169 independent constraints on the deletion.

Established. The expression strand sits at 0.241 against a ceiling of 0.775. The round produced no valid comparison between mechanisms, because every arm was scored before saturation. It produced three things that hold: the additive operator provably cannot express a pair term; loss and metric diverge on this target and not on 010; and the one large available signal is imputation from other genes of the same strain, measured to be orthogonal to the genotype-only task.

Cannot say. Whether any of the pair-rank mechanisms help. Whether the graph prior is the right prior for this task. Whether a different objective closes the loss-metric gap.

3 Morphology ✗

3.1 The target ✗

The **morphology strand** predicts the CalMorph phenotype vector of a single-deletion strain. CalMorph's nominal vocabulary is 501 parameters, which resolve into 281 base parameters and 220 coefficient-of-variation statistics; the served target is the 281 base parameters, of which the model scores 278 after three near-degenerate features are dropped. Source is Ohya 2005 (doi:10.1073/pnas.0509436102), 4,718 deletion mutants plus 122 independent *his3* wild-type replicates.

Raw CalMorph values span about eight orders of magnitude, from 6.7×10^{-5} to 1.49×10^{4} in absolute mean, so an unnormalized mean-squared-error loss is dominated by whole-cell size and actin-region size and read ≈ 4.7 million. The training path now applies a per-feature z-score fit on the training split only, which brings the loss to order 1 and makes "predict each feature's mean" score zero rather than win. Ohya's own normalization is a per-parameter Box-Cox fit on wild-type data followed by standardization, and it discards 247 of 501 parameters on a Shapiro-Wilk normality test at $p \geq 0.5$. That verdict is a normality test over base and CV parameters together, not a variance floor on the 278 modeled here, so it was not imported.

3.2 The ceiling was never measured until now, and it is high ✗

Ohya gives one biological sample per mutant but 122 wild-type replicates, so per-feature reliability is estimable as $1 - \sigma^2_{\mathrm{WT}}/\sigma^2_{\mathrm{mutants}}$ and the achievable Pearson is its square root.

quantity	value
mutants	4,718
wild-type replicates	122
modeled features	278
mean reliability	0.444
mean ceiling over modeled features	**0.611**
median ceiling	0.647
features with ceiling > 0.5	201 of 278 (72 %)
best observed (`vsceij2v`, peak epoch 27 of 65)	**0.0824**
fraction of ceiling realized	**13.5 %**

Table 8: Morphology has the second-highest ceiling of any strand here and the lowest fraction of it realized. Nearly three quarters of the modeled features are individually measurable to $r > 0.5$.

3.3 The strand was never really run, on two counts ✗

Twenty-three runs exist in `torchcell_019_morph_v5`, spanning 11 to 121 epochs, one of them a collapsed constant predictor. The best peaks at epoch 27 of 65. The joint expression-plus-morphology project is thinner still: 32 runs, none past 87 epochs, best `val/mean/pearson_per_feature` of 0.0505.

Set against section 2, where the same architecture on a related target was still improving at epoch 9,188, a 65-epoch morphology run carries no information about whether morphology is learnable. The morphology result is not a null. It is an absence of measurement, and the two must not be recorded the same way.

And the training set was a quarter of what exists, in every project. This was checked against all eight projects that carry a morphology head rather than against one, because the first version of this claim rested on `torchcell_019_morph_v5` alone and the natural objection is that some other round did train on the full screen. None did.

group	projects	runs	n_{train} observed
morphology alone (`morph_v2`, `_v3`, `_v5`)	3	183	1,161
morphology with expression (`expr_morph` .. `_v5`)	5	214	1,161
total	**8**	**397**	**1,161, no other value**

Table 9: Every morphology run ever launched, across all eight projects that carry a morphology head, trained on the same 1,161 strains. The Ohya screen holds 4,718 deletions and 1,440 of them also carry expression; the configs set `require_modalities:[expression_log2_ratio,calmorph]`, which pins every arm to that intersection, and the 80/20 split takes it to 1,161. For contrast, the same census shows the expression strand's training set growing across rounds (1,125 to 1,253) and the betaxanthin rounds differing (3,694 to 4,235), so a constant is a finding here rather than an artifact of how the census reads.

SRC: `experiments/019-simb-multimodal/results/project_census.json`

SRC: `experiments/019-simb-multimodal/conf/delta_joint_expr_morph_000.yaml`

SRC: `experiments/019-simb-multimodal/results/fig3_overlap_census.json`

The claim does not rest on the table alone. `require_modalities` is set in the config that every morphology arm inherits, and the overlap census independently gives `ohya_with_expression` = 1,440 against `ohya_unique` = 4,718. The W&B column is corroboration of a restriction that is visible in the configuration.

That restriction is correct *for the auxiliary-task question*, since comparing a joint model against a morphology-only model on different instance sets would confound the auxiliary head with a data-quantity difference. It is wrong for the question "how well can morphology be predicted", and the same configs were used for both. **Morphology has never been trained on its own full dataset.**

3.4 Which second label should morphology be paired with ✗

The overlap census answers this directly, and the answer is not expression.

genotypes carrying	count
morphology (Ohya, total)	4,718
morphology and fitness	**4,220**
morphology and expression	1,440
morphology, fitness and expression	1,326

Table 10: Fitness shares nearly three times as many strains with morphology as expression does, and covers 89 % of the morphology screen against expression's 31 %.

Pairing morphology with fitness keeps 4,220 of the 4,718 strains, so the joint arm and the morphology-only arm can be run on a large common instance set without the restriction above costing three quarters of the data. Pairing it with expression caps both arms at 1,440.

This does not retire the expression-helps-morphology question. That question is what `delta_joint_expr_morph_000.yaml` was built to answer, its control is right, and its three conditions (`expr`, `morph`, `expr_morph`) are the correct decomposition. It has simply never been run past 87 epochs. What the census changes is the order: the fitness pairing is where the statistical power is, and it can be run at full scale first.

What the earlier "0.05 to 0.14 rise" was. A flatten-Pearson computed over the $[B, 281]$ matrix before per-feature normalization, which is dominated by between-feature scale: large features are large in both prediction and target. After z-scoring, the same smoke run reports ≈ 0 on that metric. The honest metric is the per-feature correlation across strains, averaged over features, which is what table 8 scores.

3.5 A scalar target is a reasonable next step, and this is which scalar ✗

Ranking single features on reliability alone selects ones that are precisely measured and nearly constant across mutants. The rank key below is ceiling multiplied by robust coefficient of variation, IQR over absolute median across the 4,718 mutants, so a feature has to be both measurable and moved by deletions.

feature	CalMorph label	ceiling	robust CV	score
`A113`	actin_n_ratio	0.873	2.369	2.068
`C124_C`	Medium_bud_ratio	0.698	0.886	0.618
`A110_A1B`	Actin_f_ratio	0.637	0.939	0.598
`C125_A1B`	Large_bud_ratio	0.784	0.630	0.494
`A105`	actin_a_ratio	0.655	0.617	0.404
`D212`	nuclear_B_ratio_to_nuclear_AA1BC_cells	0.604	0.642	0.387

Table 11: Top of the scalar-target shortlist. The list is dominated by bounded ratios (actin localization class, bud-size class, nuclear-stage class), which is what a deletion actually moves. Size features are the opposite trade: `C115_A` whole-cell axis ratio has a ceiling of 0.972 and almost no spread across mutants.

On "predict colony size first". Colony size is fitness, and fitness is already the strongest strand in the project, trained on the Costanzo and Kuzmin single-mutant fitness data. Using it as the morphology warm-up would test the pipeline rather than the phenotype. Within CalMorph the equivalent of a colony-size scalar is a cell-size feature, and those are exactly the features deletions barely move. **Recommendation: run the scalar warm-up on `A113` or `C124_C`**, both of which are reliable and genuinely variable, and treat the run as a convergence-budget calibration for the vector target rather than as a result.

Established. Morphology has a mean ceiling of 0.611 over 278 features with 201 of them individually above 0.5, and the best score achieved is 0.0824, at 13.5 % of it. That score comes from a 1,161-strain training set drawn from a 4,718-strain screen. Target normalization is settled and the earlier apparent rise was a scale artifact.

Cannot say. Anything about whether morphology is learnable from genotype, because no morphology run has passed 121 epochs or seen more than a quarter of the data. Anything about whether expression and morphology share structure, because the joint project's longest run is 87 epochs.

4 Betaxanthin production ✗

4.1 The task, its ceiling, and the two names that get confused ✗

Cachera is a genome-wide betaxanthin deletion screen (doi:10.1093/nar/gkad656): the four-gene betalain cassette *CYP76AD1*, *DOD*, *ARO4*$^{\mathrm{K229L}}$ and *ARO7*$^{\mathrm{G141S}}$ was transferred by CRI-SPA into each strain of the yeast knockout collection and per-colony fluorescence read as a production proxy. It supplies 4,735 single deletions and is the ground truth here. **Flux Cone Learning** (FCL) is Merzbacher et al.'s method, which learns from flux samples drawn through a genome-scale metabolic model; `RandomForestClassifier_Resampled` is its best variant. The comparison is CGT against FCL, both scored on the Cachera screen.

Betaxanthin is the only phenotype in this document with a high ceiling. Its records carry a per-strain standard error over a median of 15 colonies, giving a reliability of 0.836 and a Pearson ceiling of **0.914**. SRC: `experiments/019-simb-multimodal/results/pigment_noise_ceiling.json`

4.2 Two betaxanthin numbers, and why the comparison uses the lower one ✗

round	n_{train}	best val r	split
`betaxanthin_002` (in `torchcell_020_betaxanthin`, `_v3`)	4,235	**0.4301**	ordinary ratio split
`020_v4` (`torchcell_020_betaxanthin_v4`)	3,698	0.372	Merzbacher test genes pinned out

Table 12: The highest betaxanthin score is 0.4301, 47 % of the 0.914 ceiling. Its top five trials span 0.0206, which is 0.68σ against a measured replicate σ of 0.0302, so it is a plateau rather than one lucky trial; it is nonetheless a maximum over 37 trials, and de-selecting that inflation (0.064) puts the floor for a configuration chosen without hindsight at ≈ 0.366.

The comparison number is lower than the best number on purpose. The head-to-head below runs on `020_v4`, whose split **pins the 640 Merzbacher test genes out of training**. That costs 537 training strains, and it costs score. Using the 0.4301 model instead would mean scoring on genes it had trained on, which is the one thing that would void the comparison. The cell carried forward is `s09_L6_maskon_lr0.0001_yj`, selected by validation at 0.3639 and never by its score on the comparison genes.

Both numbers are real and they answer different questions: 0.4301 is what the model does on this screen, 0.372 is what it does with a competitor's test set held out. Quoting the first in a sentence about the comparison would be the error; quoting the second as "our betaxanthin result" understates it.

4.3 The published accuracy comparison is empty for both methods ✗

FCL reports gene-level accuracy 0.700 against a majority-class rate of 0.673, achieved by calling 607 of 640 genes medium, which is 94.8 %. Our absolute binning sits at 0.681 doing the same thing. Forced to the true class marginal, FCL drops to 0.556 and CGT to 0.549.

The structural point. Every model that finds more high producers has worse overall accuracy, because the only way to call more highs is to stop calling everything medium. Accuracy selects against the capability the task is about, so it should not be the primary metric in any writeup, including ours.

4.4 Both methods have real signal, and it lives in the tail ✗

Of the top k genes each method nominates, the fraction that are truly high producers, against a base rate of 0.156:

model	$k = 10$	$k = 25$	$k = 50$
FCL RF	8 (5.1×, $p = 10^{-5}$)	19 (4.9×, $p = 8 \times 10^{-12}$)	20 (2.6×, $p = 10^{-5}$)
CGT	**9** (5.8×, $p = 4 \times 10^{-7}$)	18 (4.6×, $p = 10^{-10}$)	**24** (3.1×, $p = 2 \times 10^{-8}$)

Table 13: Precision at k for high producers. Nine of CGT's top ten are genuine high producers. Both curves approach the base rate by $k = 150$.

Median percentile rank by true class is flat for both methods, at 54/49/56 for FCL and 48/51/52 for CGT across low, medium and high. Read with the table above, that says the signal is real and confined to the tail, not that there is none.

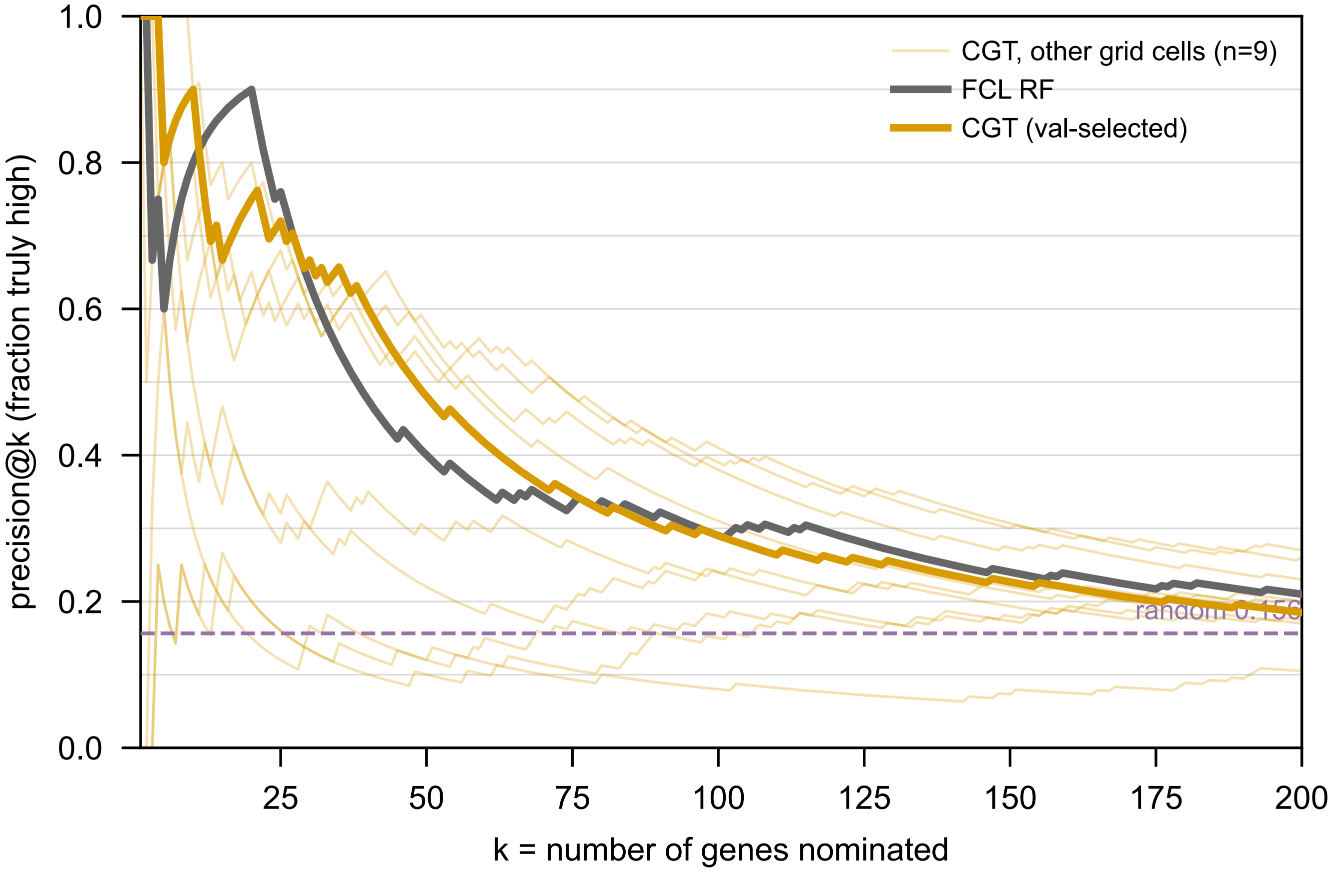
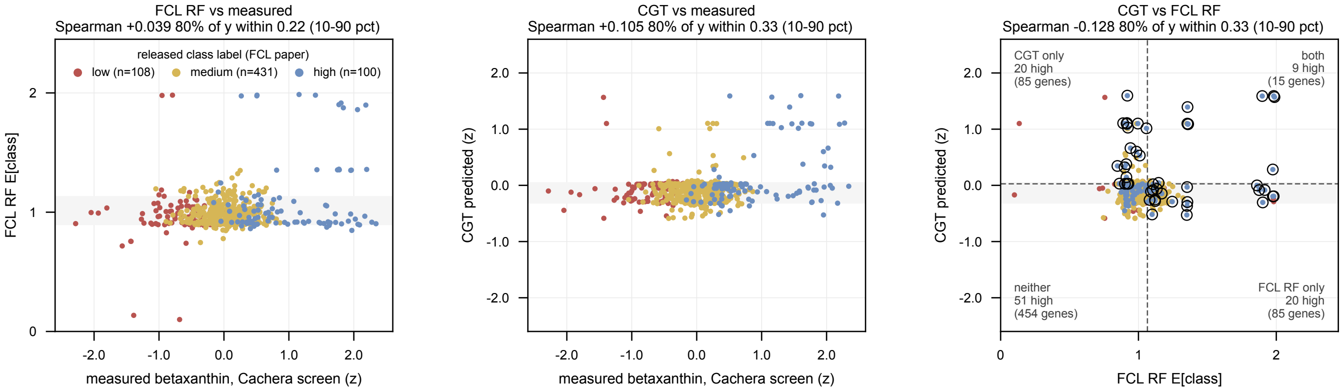


Figure 3: Precision at k for high producers, both methods, against the base rate of 0.156. Nine of CGT's top ten are genuine high producers, and both curves reach the base rate by $k = 150$. The nine unselected CGT grid cells span 0.10 to 0.65 at $k = 50$, so cell choice matters more than the gap between methods.



shaded band = middle 80% (10th-90th percentile) of the y-axis model's own output; a narrow band means the model returns nearly the same value for every gene.
Panel 3 quadrants are cut at each model's rank-matched high threshold; counts are TRUE HIGH producers (blue) with the quadrant's all-gene total in parentheses. Rings = the 49 of 100 true highs found by at least one method (FCL 29, CGT 29, shared 9).
Color is the FCL paper's RELEASED class label; x is our copy of the Cachera screen. They agree at Spearman 0.73 (88% of attainable) and 18.8% of genes fall in a different bin, which is why the colors overlap along x -- see Fig 7.

Figure 4: Per-gene spread over the 639 shared genes. Both models emit a nearly constant value: 80 % of FCL's genes fall inside an expected-class band of 0.22 on a 0 to 2 scale, 80 % of CGT's inside 0.33 z-units. Panel c shows the disagreement concretely, with each method recovering 29 of the 100 true high producers and sharing 9.

4.5 The two methods find different genes ✗

Rank-matched into the same 100 high slots, FCL recovers 29 of the 100 true high producers and CGT recovers 29, and they share 9. The union reaches 49. Their rank correlation is -0.128, which at $n = 639$ is $z = -3.2$, where two models correlating with the truth at $+0.105$ and $+0.039$ would through the truth alone correlate with each other at about $+0.004$.

No mechanism is claimed. for the sign. What is claimed is the operational consequence: recall is near-additive because the two methods are finding largely different genes, which is an argument for running both, and which is invisible in every aggregate statistic. The 51 that neither finds is the honest ceiling on this comparison.

4.6 The real difference is coverage ✗

yeast-GEM contains 1,161 genes, of which 915 appear in the Cachera screen. That is 19.4 % of its 4,721 deletions. A flux-based method has features for those and for nothing else.

population	in yeast-GEM	outside
every deletion in the screen	915 (19 %)	**3,806 (81 %)**
top 100 measured producers	32 (32 %)	**68 (68 %)**
all high producers	60 (27 %)	**163 (73 %)**

Table 14: Metabolic genes are enriched among high producers, at 26.9 % of the highs against 19.4 % of the screen, a 1.4× enrichment that is exactly what the biology predicts. It is not enough to matter: restricting to metabolism buys that concentration and costs three quarters of the hits.

On held-out test genes split by yeast-GEM membership, with identical derived labels on both sides, CGT's top 25 recovers 17 of 48 true highs inside the model (9.3× over base rate, $p = 2 \times 10^{-15}$) and 9 of 21 outside it (4.4×, $p = 2 \times 10^{-5}$).

FCL's prediction for the outside set is not poor, it is undefined: flux sampling yields no feature for a deletion the metabolic model does not contain. The FCL test set is 639 of 640 inside yeast-GEM; of our other test genes only 21 of 294 are.

Two qualifications that must travel with this claim. The rank correlation reads higher outside the model than inside ($+0.270$ against $+0.124$), but the two are measured on different, non-overlapping gene sets and a Fisher r-to-z gives $z = 2.07$, $p = 0.039$. Read it as at least as good outside as inside. And labels for the non-GEM genes are derived by us, since the FCL paper never labeled them, so the comparison to make is panel against panel.

4.7 Where the comparison is soft ✗

- **The selection is asymmetric, and it favors FCL.** Ours is the validation-selected cell. Theirs is the best of four models, justified by a test-set number, and their validation folds appear to be drawn from the same 640 genes

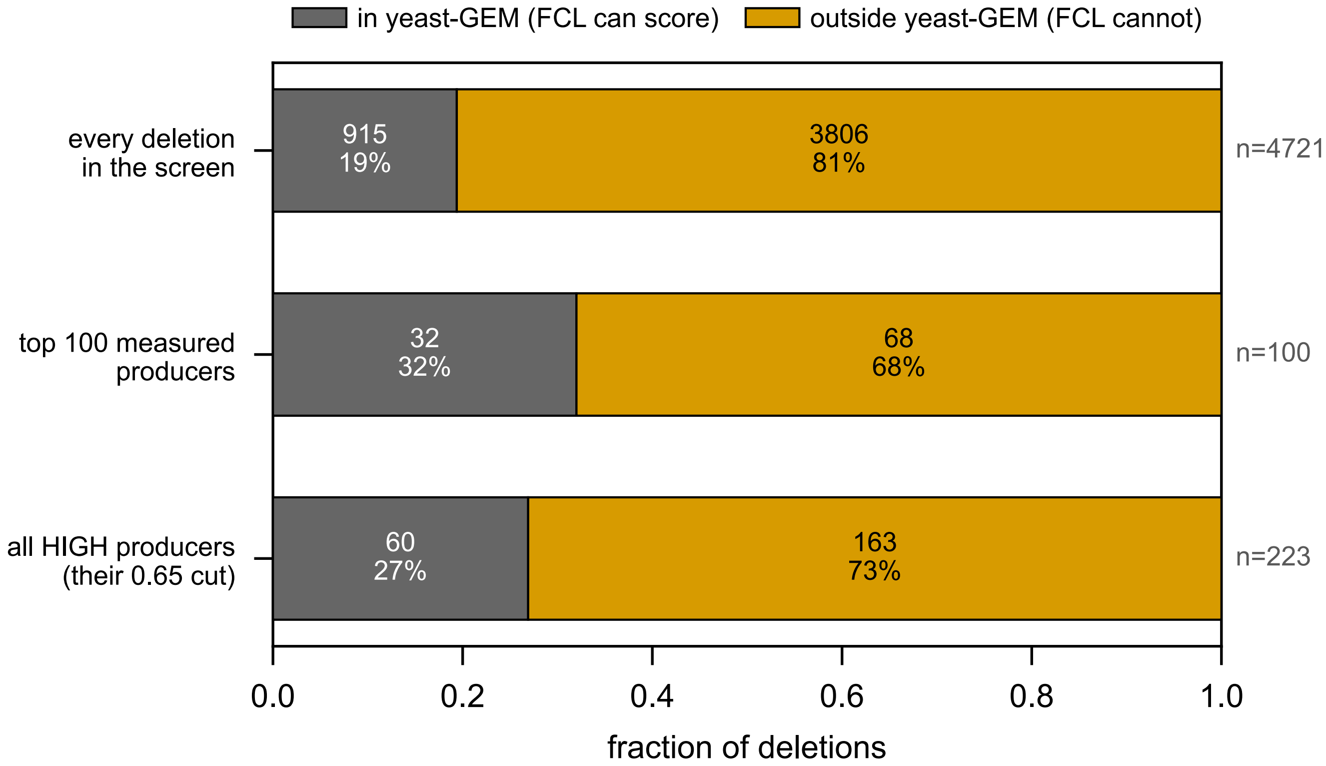


Figure 5: Measured data only, no model and no prediction: how little of the betaxanthin screen a flux-based method can reach.

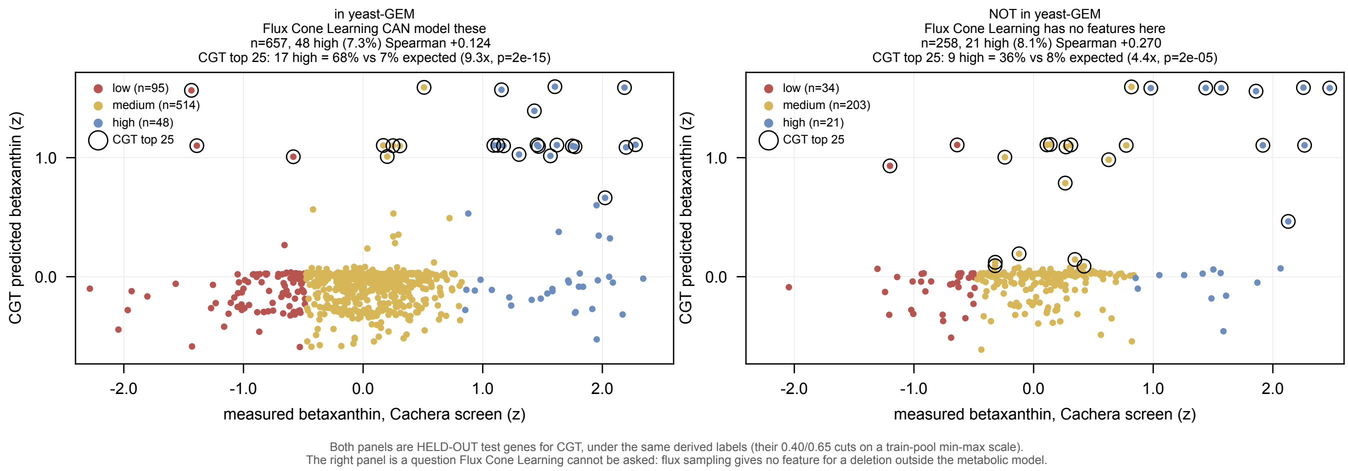


Figure 6: CGT predicts high betaxanthin producers among deletions a flux-based method cannot represent. Held-out test genes split by yeast-GEM membership, identical derived labels on both sides, circled points are CGT's top 25. Flux Cone Learning has no prediction for the right panel.

as their test list. Where the two land level, we are level with an opponent picked with more information than we allowed ourselves.

- **Validation Pearson does not track test AUC.** Four of ten CGT cells beat FCL's AUC of 0.570 and the best reaches 0.695, while the validation-selected cell reads 0.557. The runner-up on validation, 0.3528 against 0.3639, has an AUC 0.14 higher. Neither "CGT's AUC is 0.695" nor "CGT's AUC is 0.557" is safe; the metric does not separate the two methods at this resolution.
- **Their labels are a headroom bound.** Their binning rule min-max scales production and cuts at 0.40 and 0.65, so applied to our larger copy of the same screen it gives 107/476/56 against their 109/431/100. 18.8 % of genes sit in a different bin than our values would give, so a model predicting our measurement perfectly is marked wrong on about a fifth of genes.
- **Regression against classification.** They tried regression and abandoned it, in their words "challenging with the limited number of knockouts at the high and low ends." Our ceiling of 0.914 makes regression available to us. The comparison is therefore between a regressor and a three-class classifier, which is why every metric above is either rank-based or computed after binning our predictions with thresholds fit on the training split only.
- **The build is stale** with respect to the gene-name resolver (issue #195), which the split builder avoids by working from the raw screen, and which the training data still inherits until a rebuild.

Established. On this screen CGT and FCL are tied where both can be measured, both have tail-confined signal at roughly 5× enrichment in the top 25, and they find different genes. CGT predicts the 81 % of the screen a flux-based method cannot represent, and finds high producers there at 4.4× over base rate.

Cannot say. That CGT beats FCL on any single aggregate. That the rank anti-correlation means anything mechanistically. That the absolute top-100 membership is a deployable call, since the rank-matched comparison hands both sides the true class marginal.

5 Amino-acid pools, and whether they help production ✗

5.1 The target ✗

Mulleder 2016 (doi:10.1016/j.cell.2016.09.007) measured the intracellular concentration of 19 amino acids in 4,678 single-deletion strains. There is one replicate per strain and no released standard error, so the dataset admits **no within-dataset reproducibility ceiling**. Every correlation involving it is attenuated by an unmeasured amount, and that qualification travels with every number below.

The best validation `pearson_per_feature` is **0.211**, run `24vlgope` in `torchcell_022_mulleder19_v4`, peaking at epoch 402 of 999. The median across the 31 runs is 0.0498, so the leaderboard is a long tail under one good configuration rather than a plateau. Earlier rounds read ≤ 0.025, then 0.171, so the trajectory is upward across rounds.

5.2 These runs converge, which the expression runs never do ✗

Twenty-four runs in the preceding round were scored for their convergence shape. The typical run peaks at epoch 48 to 134, continues for another 41 epochs with validation slope turning negative while training slope stays positive, and is classified converged then overfitting. None of them hit the epoch cap. SRC: `experiments/022-mulleder-metabolome/results/training_length_analysis.json`

This is the opposite failure to section 2, where nothing peaked at any budget up to 10,000 epochs. The two strands run the same architecture on the same encoder, so whatever makes expression climb for thousands of epochs is a property of the expression target or its loss, not of the model.

5.3 The precursor hypothesis is wrong, and the profile hypothesis is right ✗

Betalains derive from L-tyrosine (tyrosine to L-DOPA to betalamic acid), and a betaxanthin is betalamic acid condensed with an amino acid. The mechanistic prediction is that a deletion's free amino-acid pool carries information about how much betaxanthin that same deletion makes. Both screens are keyed by systematic ORF and share 4,432 deletions, so the prediction is directly testable with no model in the loop.

What the prediction actually is, since it is easy to read as a model result and is not one. No neural network appears anywhere in this subsection. The procedure is:

1. Join the two screens on systematic ORF. One row per deletion: 19 measured amino-acid concentrations from Mulleder, and one betaxanthin value from Cachera.
2. Take log of each concentration (they are skewed positive quantities) and z-score each of the 19 columns.
3. Fit **ridge regression** from those 19 numbers to the betaxanthin value, with the penalty chosen by inner cross-validation.
4. Score **out of fold**: five-fold cross-validation, so every prediction is made by a model that never saw that deletion, then correlate the held-out predictions against the measurements. Repeat over three fold shuffles and report the mean.

So $r = 0.298$ means: *knowing a deletion's 19 amino-acid concentrations lets you predict its betaxanthin at* $r = 0.298$ *on deletions you did not fit on.* The comparison row "tyrosine alone" is the same procedure with one column instead of nineteen. It is a statement about the two measurements, and it puts a floor under what any model sharing a representation between these phenotypes could hope to exploit.

predictor of betaxanthin	out-of-fold Pearson r	n
tyrosine alone (the precursor)	0.064	4,432
19 amino acids jointly	**0.298**	4,432
on the 3,708 deletions with a Costanzo fitness record		
single-mutant fitness alone	0.151	3,708
19 amino acids	0.176	3,708
19 amino acids and fitness	0.200	3,708
19 amino acids, fitness regressed out of both	0.133	3,708
on the 724 with no Costanzo record		
19 amino acids	**0.455**	724

Table 15: Cross-validated ridge, five folds, three shuffles; the spread across shuffles is ≤ 0.022 everywhere and ≤ 0.004 on the large sets. Amino acids enter as log concentration, z-scored.

Tyrosine is not the axis. Its marginal correlation with betaxanthin is $r = -0.076$ ($p = 3.9 \times 10^{-7}$), which is negative, and it ranks sixth of nineteen by absolute correlation. The strongest is methionine at $+0.140$, followed by aspartate $+0.127$ and arginine -0.118. No amino acid exceeds $|r| = 0.15$. Correcting for the betaxanthin side's known reliability of 0.836 raises these by a factor of 1.094, which does not change the reading; the Mulleder side's correction is unmeasurable and would raise them further by an unknown amount.

The profile is the axis. The 19-dimensional pool predicts betaxanthin at out-of-fold $r = 0.298$, roughly 4.7× the precursor alone. For scale, the CGT's own genotype-to-betaxanthin validation Pearson is 0.372 to 0.430, so a linear map from 19 measured metabolite concentrations recovers most of what the model recovers from genotype.

Why fitness enters at all, since it is a third dataset and looks like a detour. Both screens are deletion phenotypes and both respond to how well the strain grows. A slow deletion can have a distorted amino-acid pool *and* a distorted pigment readout with no metabolic relationship between the two, which would produce exactly the correlation above for an uninteresting reason. Costanzo single-mutant fitness (KanMX deletions, 30 C) is the standard measurement of that shared nuisance, so it is regressed out of both sides and the fit recomputed on the residuals. It is a control, not a third phenotype in the story.

Single-mutant fitness alone predicts betaxanthin at 0.151, so the shared growth component is real and substantial. Regressing it out of both sides leaves the amino-acid profile at 0.133 against 0.176 without the control on the same genes, so roughly three quarters of its predictive power survives. Combining the two reaches 0.200, above either alone.

The control also shrinks the gene set, from 4,432 to the 3,708 deletions that carry a Costanzo record, which is why both the controlled and uncontrolled fits are reported on that same 3,708 rather than compared across different populations.

And most of the signal is where fitness is unavailable. The 724 deletions with no Costanzo KanMX record have a betaxanthin standard deviation of 0.914 against 0.464 for the rest, and on them the amino-acid profile predicts at 0.455. This is the population that carries the extremes, and it is the population a fitness-based control cannot reach.

A mechanism, labeled as unverified. The Cachera background carries $ARO4^{\mathrm{K229L}}$ and $ARO7^{\mathrm{G141S}}$, feedback-resistant alleles that release the shikimate pathway from the control that shapes the pools Mulleder measured in the plain deletion collection. A plausible reading is that free tyrosine in an unengineered strain is not the quantity that limits flux in an engineered one. This has not been tested and no experiment here can test it.

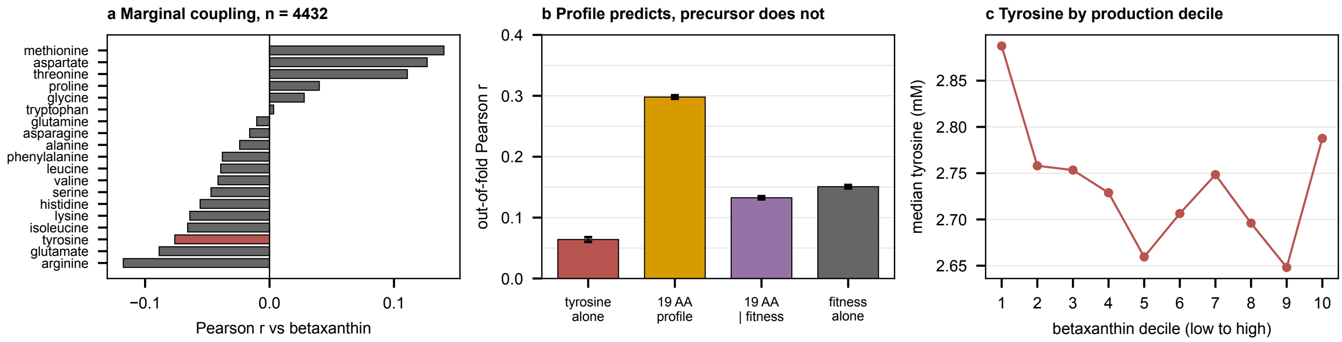


Figure 7: (a) Marginal correlation of each amino acid with betaxanthin over the 4,432 shared deletions; tyrosine highlighted. These reproduce `experiments/019-simb-multimodal/results/pigment_noise_ceiling.json` exactly, which the generating script asserts rather than assumes. (b) Cross-validated ridge fits from table 15. (c) Median tyrosine by betaxanthin decile. The relationship is non-monotone: tyrosine is highest in the lowest-production decile, falls through the middle, and rises again at the top, which is why a linear correlation reads near zero. The axis spans 2.65 to 2.89 mM, so the swing is about 9 % of the median.

5.4 Does the metabolome head actually help the betaxanthin head ✗

The claim under test is narrow and was stated before the round: in Delta grid cell `s04_L6_maskon_lr0.0001`, the joint arm read +0.0203 over the control on `val/betaxanthin/pearson_per_feature` at a matched budget of 333 epochs. That was one paired difference at one seed with no estimate of the noise floor on it. The same grid read −0.049 and −0.060 in the two $L = 2$ cells, both run to 1,000 epochs, so the sign flips with depth.

grid cell	control	joint	Δ	aux r	epochs (ctrl/joint)
`s00_L2_maskon_lr0.0001`	0.3927	0.3473	-0.0455	0.149	999/999
`s02_L2_maskoff_lr0.0001`	0.3570	0.3078	-0.0492	0.053	999/999
`s04_L6_maskon_lr0.0001`	0.3769	0.3960	**+0.0192**	0.113	999/999
`s06_L6_maskoff_lr0.0001`	0.4216	0.3609	-0.0608	0.061	999/999
`s08_L4_maskon_lr0.0001`†	0.4105	0.4199	**+0.0094**	0.120	999/747
`s10_L4_maskoff_lr0.0001`	0.3705	0.3745	**+0.0039**	0.054	999/999
`s01_L2_maskon_lr0.001`*	0.0553	0.0399	-0.0155	0.029	999/999
`s03_L2_maskoff_lr0.001`$^{\dagger *}$	0.0400	0.0474	**+0.0074**	0.013	490/999
`s05_L6_maskon_lr0.001`*	0.0674	0.0126	-0.0548	0.012	999/999
`s07_L6_maskoff_lr0.001`*	0.0699	0.0587	-0.0112	0.009	999/999
`s09_L4_maskon_lr0.001`*	0.0461	0.0529	**+0.0068**	0.009	999/999
`s11_L4_maskoff_lr0.001`*	0.0611	0.0478	-0.0133	0.010	999/999
mean, cells where the control learned (5 cells)			-0.0265 ± 0.0159		2 of 5 positive
mean, all matched-exposure cells (10 cells)			-0.0220 ± 0.0090		3 of 10 positive

Table 16: Betaxanthin score with and without a 19-amino-acid auxiliary head, paired within grid cell, on `val/betaxanthin/pearson_per_feature`. Δ is joint minus control. *aux* is the auxiliary head's own score, carried because an auxiliary head at $r \approx 0$ that still moved the primary metric would mean the gain came from regularization rather than from shared metabolic signal, and those are different findings. † marks a cell whose two arms ran epoch counts differing by more than 50; since the score is a running maximum, such a difference is part effect and part unequal exposure, and those rows are excluded from both means. * marks a cell whose control arm scored below 0.10, meaning neither arm learned the task; every one of them is an lr $= 10^{-3}$ cell, and a difference between two failed runs measures nothing about the auxiliary head.

Scored uniformly across the whole grid, the effect is **negative**. Over the five matched-exposure cells whose control arm learned the task at all, the mean difference is $\mathbf{-0.0265 \pm 0.0159}$ (standard error, $n = 5$), with 2 of 5 positive. Over all ten matched-exposure cells including the collapsed ones it is -0.0220 ± 0.0090, with 3 of 10 positive.

Three readings the table supports, in decreasing confidence.

1. **The original** +0.0203 **reproduces, and it is one cell.** Cell `s04_L6_maskon_lr0.0001` reads +0.0192 here under a different scoring rule and a different epoch budget, which is close agreement. It is also the only clearly positive live cell, against two large negatives at $L = 2$ and one at $L = 6$ with the graph mask off.
2. **Depth and the graph mask both matter more than the auxiliary head.** The two $L = 2$ cells read −0.0455 and −0.0492; the two $L = 6$ cells read +0.0192 with the mask on and −0.0608 with it off. Any single-cell claim about the metabolome head is smaller than the spread across the cells it could have been made in.

3. **The auxiliary head is weak wherever the primary is good.** Its own score runs 0.053 to 0.149 across the live cells, well under the 0.211 the amino-acid strand reaches when it is the only head. So the joint arm is paying capacity for a head that is itself underperforming, which is a different problem from the coupling being absent.

The measurement this still lacks. Every number in the table is $n = 1$ per arm per cell except `s08`, so the spread across cells mixes the effect with initialization noise, and no within-cell noise floor exists. That is exactly the gap the replication config was written to close.

The designed replication, `experiments/023-metabolome-betaxanthin-joint/conf/igb_bx_aa.yaml`, adds a third setting: an auxiliary head restricted to **tyrosine only**, against a common control, on a gene set pinned by `require_head_targets` so no arm can win by seeing more rows. **It has not been run**; no W&B project `torchcell_023_bx_aa_v1` exists. Given table 15, its tyrosine arm is now predicted to be the null one and its 19-amino-acid arm the live one, which is a sharper prediction than the config was written against.

Established. The free amino-acid profile of a deletion predicts that deletion's betaxanthin production at $r = 0.298$ out of fold, tyrosine alone predicts it at 0.064, and about three quarters of the profile's power survives regressing out single-mutant fitness. The amino-acid strand itself reaches 0.211 and converges rather than under-trains.

Scored across the whole 023 grid, adding the metabolome head *lowers* betaxanthin by -0.0265 ± 0.0159 over five comparable cells. The one positive cell is the one the original claim came from.

Cannot say. How much of the 0.298 is attenuated by Mulleder's unmeasurable noise, so it is a floor rather than an estimate. Whether a shared encoder can exploit the coupling: the data says the coupling is there, the joint training says the current head does not use it, and at $n = 1$ per cell those two are not yet in contradiction.

The gap this exposes. A linear map on 19 measured metabolite concentrations reaches 0.298 on betaxanthin, and the auxiliary head that is supposed to inject the same information reaches 0.053 to 0.149 on those metabolites and then costs the primary task 0.027. The coupling is not the bottleneck. The route by which a second head's representation reaches the first head's readout is.

6 Beta-carotene production ✗

6.1 Why this target is different from the other pigment ✗

Ozaydin 2013 (doi:10.1016/j.ymben.2012.07.010) screened the deletion collection carrying a beta-carotene pathway and scored each colony's color **by eye on an ordinal scale from -5 to $+5$**. The dataset holds 4,474 records, of which 4,344 have a single replicate and 130 have two or three. That is a different kind of measurement from the Cachera fluorescence readout, and it needs a different ceiling: a Pearson ceiling on a subjective ordinal is the wrong object, so rank agreement is used.

Two estimates exist, and they disagree by a factor of seven.

estimate	basis	n	Spearman
replicate max against min	130 replicated rows	130	**0.544**
independent re-screen	`2ndRoundOfTransformations` sheet	119	0.075
range-restriction corrected	Thorndike case 2, Pearson	119	0.105

Table 17: Reliability of the beta-carotene score. Neither estimate is clean. The max-against-min comparison is biased in both directions at once: the two are drawn from one replicate set, which inflates agreement, and they are the widest possible split of that set, which deflates it. The re-screen is a genuine test-retest but was run on selected top hits, so its 1st-screen scores sit almost entirely in $3 \ldots 5$ of a $-5 \ldots +5$ scale.

The reported ceiling is 0.544, from the replicate comparison. Read the 0.075 as what a re-screen of selected winners looks like under range restriction, not as evidence that the score is worthless.

6.2 What was measured ✗

Thirty-nine runs in `torchcell_021_beta_carotene_v4`, spanning 134 to 999 epochs. The best validation `spearman_per_feature` is **0.132**, run `o0aadmot`, peaking at epoch 799 of 999. Median across live runs is 0.072. Against the 0.544 ceiling that is 24 % realized.

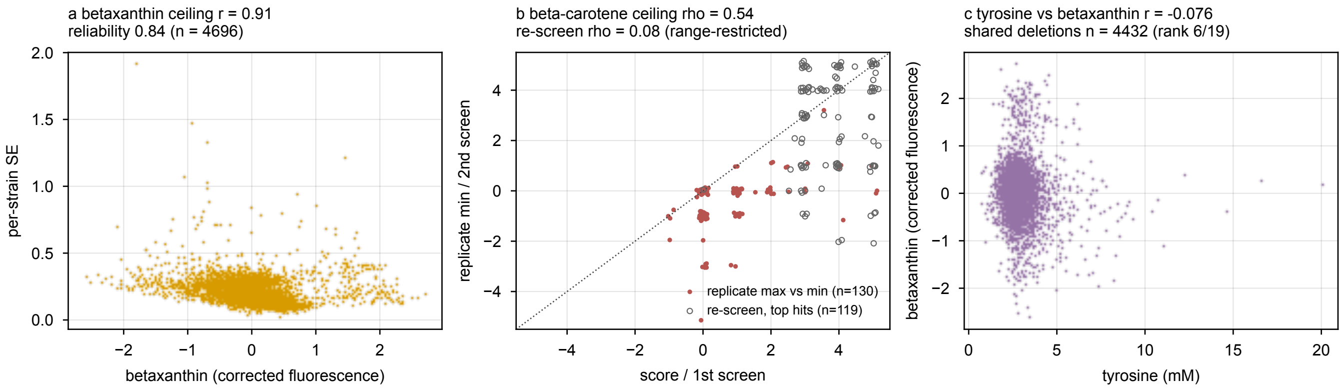


Figure 8: Reliability of both pigment targets. The betaxanthin panel uses the per-record standard error over a median of 15 colonies; the beta-carotene panels use rank agreement, because the target is a hand-scored ordinal.

Established. Beta-carotene is the weakest pigment target and the reason is visible in the measurement rather than in the model: a hand-scored ordinal with one replicate per strain has a reliability that no architecture recovers. The strand does carry signal, at 0.132 against a chance level of zero, and its best run peaks late (epoch 799 of 999), so it shares the expression strand's under-training rather than the metabolite strands' early overfitting.

Cannot say. Which of the two ceiling estimates is right, and therefore whether 0.132 is a quarter of achievable or most of it. That gap is resolvable only with a replicate design the source data does not contain.

Recommendation. Keep beta-carotene as a second production target for the coverage argument in section 4, where the claim is about which genes a flux model can represent and does not depend on the absolute score. Do not spend a sweep on lifting its number.

7 What the strands have in common ✗

7.1 What the leaderboard covers, and what it does not ✗

The strands span **28 W&B projects and 2,187 runs**. Every one of them is enumerated, with its run count, its metric names and its training-set size, by a census that reads summaries only and therefore covers everything. SRC: `experiments/019-simb-multimodal/scripts/project_census.py`

strand	projects	runs	n_{train} values seen
expression	9	1,369	1,125 – 1,253
expression + morphology joint	5	214	1,161
morphology	3	183	1,161
betaxanthin	4	142	3,694 – 4,235
beta-carotene	3	136	3,846, 3,849
amino acid	3	97	4,234, 4,235
betaxanthin + metabolome	1	46	3,832
total	**28**	**2,187**	

Table 18: The whole account. Two columns carry findings on their own. Morphology is the only strand whose training-set size never varies, and the value it never varies from is a quarter of its screen. The betaxanthin range is why its two headline numbers are not comparable (table 12).

Scoring is a different matter. Finding a run's *best* value needs its full validation history, which is one request per run. Coverage there is not automatically complete, so it is stated before the table rather than after it, and the split below is generated from the data rather than written by hand: a hand-kept list would go stale silently, which is the worst failure mode for a table whose job is to state a limitation.

What this does and does not put at risk. A strand maximum can only move *upward* when the pending projects land, never down, so every "X of ceiling" figure below is a lower bound. The medians are the vulnerable statistic, since they are computed over the projects read.

One resolution caveat, because it is a real inconsistency. History is sampled, and two passes used different sample counts: the runs read first at 2,000 points per curve, the rest at 500. Coarser sampling can only lower a

status	projects	which
history read	27	`expr`, `expr_morph`, `expr_morph_v2`, `expr_morph_v3`, `expr_morph_v4`, `expr_morph_v5`, `expr_v2`, `expr_v3`, `expr_v5`, `expr_v6`, `expr_v7`, `expr_v8`, `expr_v9`, `morph_v2`, `morph_v3`, `morph_v5`, `020_beta_carotene`, `020_beta_carotene_v3`, `020_betaxanthin`, `020_betaxanthin_v3`, `020_betaxanthin_v4`, `020_metabolism`, `020_mulleder19`, `020_mulleder19_v3`, `021_beta_carotene_v4`, `022_mulleder19_v4`, `023_bx_m19_v1`
history pending	1	`019-simb-multimodal_cgt_multitask`

Table 19: Finding a run's best score needs its full validation history, one request per run. Against a rate-limited API that does not complete over 2,187 runs in one sitting, so the pull caches per project and resumes, and the remaining projects are queued rather than dropped. A strand maximum can only move UPWARD as they land, so every *of ceiling* figure is a lower bound. **1 still pending.**

`roll_max`, by a few thousandths in practice (the best expression run reads 0.2407 at 2,000 points and 0.2382 at 500). Where a run appears in both passes the finer one is kept, so the effect is confined to runs read only in the second pass and is downward.

7.2 One table, read the same way ✗

strand	ceiling	best	of ceiling	epochs (peak/last)	runs
Expression (Kemmeren, Sameith)	0.775	**0.229**	30%	4007/4105	857
Expression, masked-label objective	0.775	**0.241**	31%	9188/9997	16
Morphology (CalMorph, Ohya)	0.611	**0.082**	13%	27/65	183 (1 flat)
Expression and morphology, joint	–	**0.059**	–	16/27	214
Betaxanthin (Cachera)	0.914	**0.401**	44%	74/124	142
Betaxanthin with metabolome head	0.914	**0.430**	47%	176/496	46
Amino-acid pools (Mulleder)	–	**0.211**	–	402/999	97
Beta-carotene (Ozaydin)	0.544	**0.132**	24%	799/999	136

Table 20: Every strand, scored the same way. *best* is `roll_max`, the maximum of a centered five-point rolling mean of the validation curve, an upward-biased order statistic whose bias grows with epochs run, which is why the epoch column travels with it. *epochs (peak/last)* gives where the best run peaked and where it stopped: a peak in the last tenth means the run was cut off while still improving. Ceilings are measured by three different estimators because the three measurement types require them; the amino-acid cell is empty because Mulleder has one replicate per strain, and no ceiling is estimable. *flat* counts runs whose validation metric never left zero. **Coverage: 27 of 28 projects, 1691 runs.** The rest are queued (table 19); a strand maximum can only move upward when they land, so every *of ceiling* figure is a lower bound. The betaxanthin row is known to be one such case: table 12 carries 0.4301 from a study in a queued project.

Three things are visible only when the strands sit together. The ceilings differ by more than a factor of two, so an unqualified r is uninterpretable across rows. The epoch budgets differ by more than a factor of a hundred, so the score column is partly a record of how long each strand was allowed to run. And the two strands closest to their ceilings, betaxanthin and amino acid, are the two with a scalar or low-dimensional target.

7.3 Two opposite convergence failures, and neither was diagnosed from a curve ✗

- **Expression never peaks.** At every budget tried, up to 10,000 epochs, the validation Pearson peaks in the last tenth of the run. Raising the budget from 1,600 to 10,000 raised the score from 0.204 to 0.241 and moved the peak with it.
- **The metabolite strands peak early and then overfit.** Amino-acid runs peak at epochs 48 to 134 with validation slope turning negative while training slope stays positive. The best betaxanthin run peaks at epoch 254 of 999.
- **Morphology was never run long enough to tell which it does.** Longest run, 121 epochs.

Same encoder, same optimizer family, opposite behavior. The difference is in the target: a 6,127-dimensional vector of $\log_2$ ratios against a scalar or a 19-vector. Any single epoch budget applied across strands is wrong for at least two of

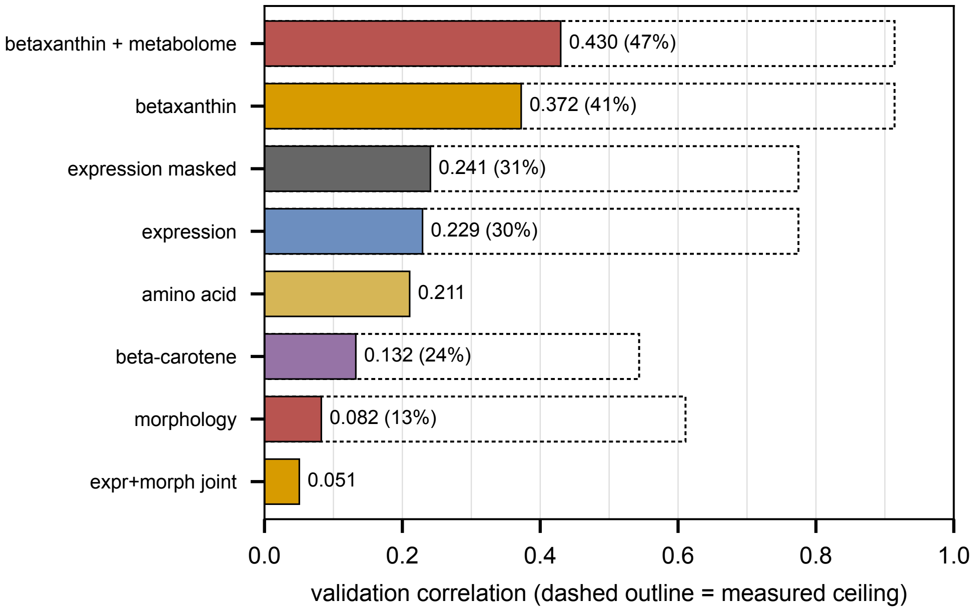


Figure 9: Achieved score against measured ceiling, best run per strand. The dashed outline is the ceiling and the gap is the headroom; amino acid and the expression-morphology joint arm carry no outline because no single ceiling applies to them. A bare correlation is not comparable across these rows, which is the reason the panel exists.

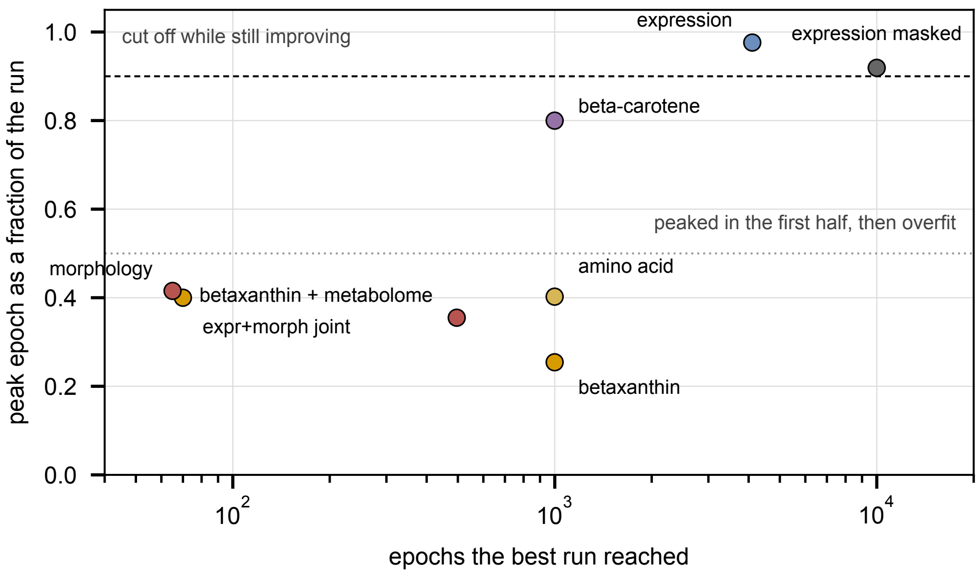


Figure 10: Where the best run's peak sits inside its own budget. Points near the top were still improving when the run stopped, so their scores are lower bounds set by compute. Points in the lower band peaked early and spent the rest of the budget overfitting.

them, which is what produced the truncated expression waves and would produce overfit metabolite runs if the budget were set from expression.

7.4 The training objective diverges from the metric on expression alone ✗

The full measurement is section 2.3; what belongs here is the cross-strand statement. On expression the *quantile loss* bottoms at epoch 463 and rises for the next 9,500 while Pearson climbs, and `nmse` never returns below 1, so the model is no better than predicting each gene's mean in squared error while ordering genes at $r = 0.236$. *Squared error itself does not diverge*: its minimum lands alongside the Pearson peak on both long runs. The 010 fitness and interaction task shows neither behavior, with both losses falling monotonically and the validation-loss minimum at the last epoch.

So the divergence is a property of one objective on one target, not a general feature of the multitask setup, and the metabolite strands, which peak early and overfit, are governed by something else. Any campaign that sets a stopping rule or a checkpoint criterion from one strand and applies it to another will be wrong for at least one of them.

7.5 Configuration choice moves results more than method choice ✗

On betaxanthin the ten grid cells span Spearman 0.013 to 0.158 and AUC 0.406 to 0.695, against a between-method gap of 0.0014 on AUC. Four low points are the $\text{lr} = 10^{-3}$ cells, which collapsed to a constant predictor. On the amino-acid strand the best run is 0.211 and the median is 0.0498.

Any single-number claim about a method is dominated by which cell is quoted, which is why the cell must be fixed by validation in advance, and why the leaderboard records the median alongside the maximum. It also means an arm difference smaller than the cell spread is not measurable without pinning the cell and replicating seeds inside it.

Is the spread just the dataset split. No, and this one has been measured rather than argued. The betaxanthin Optuna study repeated some hyperparameter configurations exactly, which gives a direct estimate of the floor:

repeated configuration	objective values	range
trials 3, 10, 11	0.4216, 0.4095, 0.3584	0.0632
trials 19, 31	0.4211, 0.3896	0.0315
pooled replicate σ (3 d.o.f.)		**0.0302**

Table 21: Re-running an identical configuration moves the objective by $\sigma \approx 0.030$. The observed spread across configurations is several times that, so configuration choice is doing real work; but any single difference under ≈ 0.03 is inside this floor.

Three components are separable in what has been run. **Collapsed configurations**: on betaxanthin and on the 023 grid every $\text{lr} = 10^{-3}$ cell fell to a near-constant predictor, and those account for the bottom of the range. **Genuine configuration effects**: depth and the graph mask move the 023 result by more than 0.05 (table 16), well outside the floor. **Run-to-run noise**: the $\sigma = 0.030$ above, which in rounds that pin `data_module.split_seed` is initialization only, since the partition is held fixed precisely so that varying `seed` varies weights and nothing else.

There is a fourth term that is not noise and is often mistaken for a result: **selection inflation**. Taking the maximum over 37 trials inflates it by an estimated 0.064, so the study's headline 0.4301 corresponds to a floor of ≈ 0.366 for a configuration chosen without hindsight. Its top five trials span 0.0206, which is 0.68σ, so they are indistinguishable from each other.

7.6 Cross-modality overlap is real and small ✗

Before spending compute on joint proteome and expression training, the linear overlap between the two was measured on the 1,350 strains they share. Per-gene correlation across strains has median $r = 0.08$, with 1.7 % of genes above 0.3 and 8.9 % negative. A ridge map recovers held-out R^2 of 0.035 in one direction and 0.024 in the other, above a trivial baseline of -0.004 but far from redundancy. SRC: `experiments/019-simb-multimodal/results/proteome_expression_eda.json`

The two modalities were measured in different media, synthetic minimal for the proteome and synthetic complete for the expression, which attenuates every number above. The overlap is therefore certified as non-redundant and not certified as cleanly complementary. That distinction is what bounds how hard a "the proteome helps expression" claim can be pushed until a same-media pair exists.

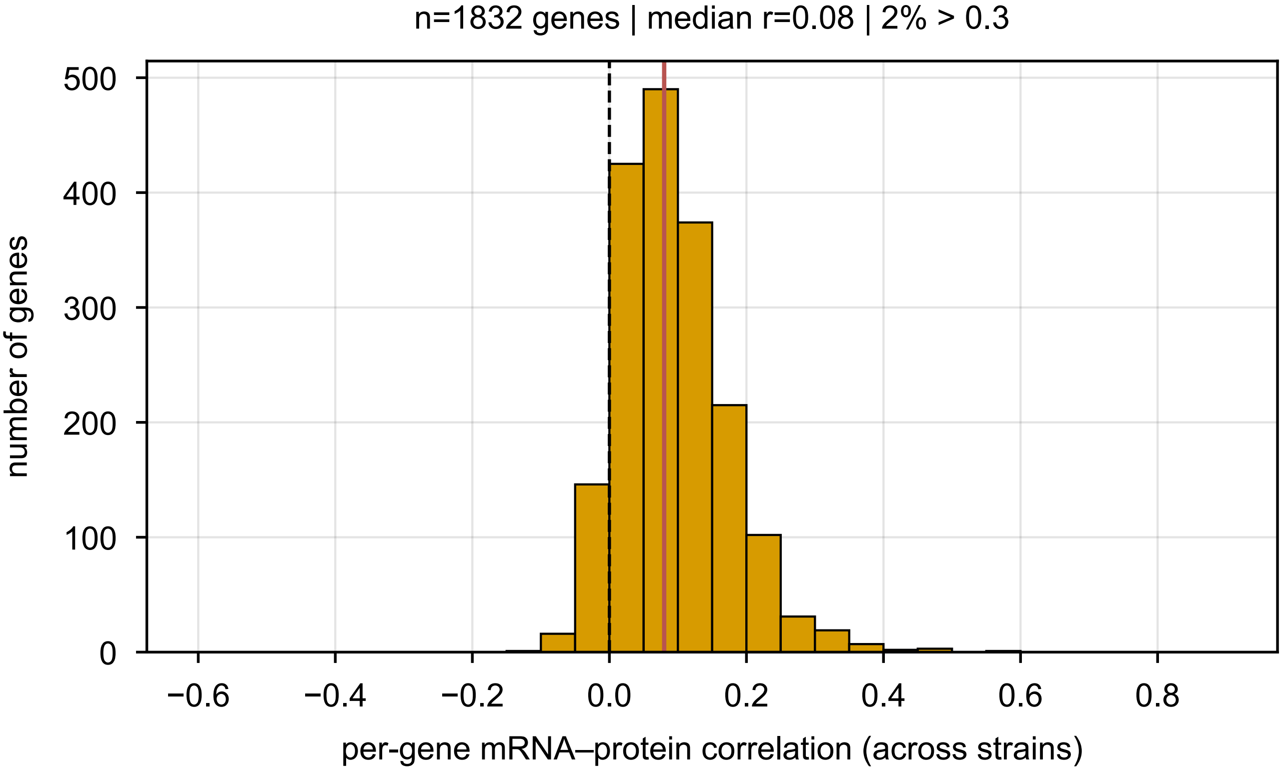


Figure 11: Per-gene correlation across strains between the Messner proteome and the Kemmeren expression compendium, over the 1,832 genes and 1,350 strains they share. A narrow lump barely off zero. The few high-r genes are dominated by the strain in which that gene is itself deleted, which drives both its transcript and its protein toward zero.

7.7 The graph prior has never been checked ✗

The nine interaction graphs enter the model as a hard additive attention mask, so composing L layers makes the influence of gene X on gene Y approximately the L-step random-walk reachability of Y from X on that graph. That is a falsifiable assumption about the data: *deleting X perturbs the genes near X on graph k.* If it is false, pulling attention toward the graph pushes the model toward the wrong prior and no setting of the regularization weight rescues it.

The probe that would test it is specified and unbuilt. It costs minutes of CPU: for each deleted gene, the across-reporter rank correlation between graph proximity and response magnitude, reported per graph as an AUC against degree-preserving rewiring and a matched random graph, where 0.5 means the graph says nothing. SRC: `notes/experiments.019-simb-multimodal.graph-prior-probe.md`

Nine graphs consume nine attention heads. If three carry signal, the other six are constrained toward noise and are worse than free heads. This is the cheapest unrun experiment in the project.

7.8 The perturbation axis is the binding resource ✗

Table 7 makes the point for expression: 9.2 million scalar labels across 1,484 perturbations, each gene deleted exactly once. Morphology is the same shape, 1.3 million labels across 4,718 perturbations, each once. The trigenic interaction data is the opposite shape, 91 thousand labels across 91 thousand records in which a given gene recurs with many partners.

The strands that work in this project are the ones where the supervision constrains a gene repeatedly. **Hypothesis, untested:** the reason multi-task training is attractive here is not that the labels are informative about each other, but that a second phenotype gives a second, differently-shaped constraint on the same perturbation. That predicts the gain should be larger where the two phenotypes share strains and smaller where they merely share genes, which is a testable statement and has not been tested.

8 What to run next ✗

Ordered by cost against what each would settle. The first four are cheap enough that not running them is the expensive choice.

8.1 Tier 0, cheap and decisive ✗

1. **The graph-prior probe** (section 7). Minutes of CPU, no GPU, no training. It asks whether deleting gene X actually perturbs X's neighbors on each of the nine graphs, which is the assumption the entire graph channel rests on and which has never been checked. Nine attention heads are currently spent enforcing it. If three graphs carry signal, regularize those three and free the other six.

2. **Morphology at full scale.** Drop `require_modalities` for the morphology-only arm and train on all 4,718 Ohya deletions instead of the 1,440 that also carry expression. This is a config change, and it is the only way the 0.0824 becomes a number about morphology rather than about a quarter of morphology. Run it long enough to see a peak, which section 7 says cannot be assumed from the expression budget.

3. **The scalar morphology warm-up**, on `A113` (ceiling 0.873, robust CV 2.37) or `C124_C` (Medium bud ratio, 0.698 and 0.886). Cheap, and it calibrates the epoch budget for the 278-feature target. Not colony size: colony size is fitness, which is already the project's strongest strand, and the CalMorph size features are the ones deletions barely move.

4. **Apply the post-hoc rescale to the existing best checkpoint.** Multiplying predictions by r/s changes no correlation and moves `nmse` from 1.010 to 0.944, taking the model from worse-than-the-mean to better-than-the-mean in squared error (section 2.3). An evening's work, no training, and it removes an objection the figure would otherwise have to answer.

5. **Bracket the expression peak, by resuming rather than restarting.** No run has observed a maximum at any budget up to 10,000 epochs, and a fresh 10,000-epoch run costs 3.8 days. Resuming the existing best checkpoint is the cheap form of the same experiment, and the epoch at which the curve turns is what sets every arm budget afterward. section 9 proposes $E = 4{,}000$ in the meantime, justified by an arm that has been observed to peak at 3,921.

8.2 Tier 1, the two questions worth GPU ✗

Both are laid out with their epoch budgets, seed counts and slot costs in section 9, which is the operational version of this subsection. In summary:

- **Settle the objective first** (table 23). The quantile loss reaches its minimum at epoch 463 and rises for the next 9,500 while Pearson climbs, and the model never gets `nmse` below 1. A mechanism arm evaluated under an objective that is optimizing something else cannot separate "the mechanism does not help" from "the objective does not reward it".
- **Then the pair term** (table 24). The reference operator provably cannot express a dependence on the pair (deleted gene, reporter gene), and masked unmasking does not supply one at the point the score is taken. A rank-64 pair term already reaches 0.2274 in 2.0 days against the masked objective's 0.2362 in 3.8.
- **Pair morphology with fitness for the first full-scale joint test**, since fitness shares 4,220 strains with morphology against expression's 1,440. Then run the expression pairing on the 1,440 as the mechanism test it was designed to be.

Substituting tyrosine for the 19 amino acids, as `igb_bx_aa.yaml` proposes, is now predicted to be the *null* arm rather than the live one, since tyrosine alone predicts betaxanthin at 0.064 against the profile's 0.298. Keep it as a negative control, not as the headline contrast.

8.3 Tier 2, the structural options ✗

Constraint-based coupling is the mechanism Figure 6 needs. The 023 result is that the coupling between amino-acid pools and betaxanthin is present in the data at $r = 0.298$ and that a plain auxiliary head does not use it, costing -0.0265 ± 0.0159 (table 16). That is a structural gap, and an unstructured second head is the wrong instrument to close it: it asks the encoder to discover a stoichiometric relationship from 19 noisy concentrations and one pigment readout, with nothing in the architecture saying the two are connected through a pathway.

A constraint-based layer says it. Penalizing predictions that violate a stoichiometric or flux balance over yeast-GEM makes the tyrosine-to-betalamic-acid route a property of the model rather than something it must infer, and it turns the metabolome head from an auxiliary target into a set of intermediates the constraint actually references. **This is the Figure 6 mechanism, and it is what makes the amino-acid panel a step in an argument instead of a negative result.**

Two facts size it rather than block it. yeast-GEM covers 19.4 % of the deletions in the screen and 27 % of its high producers (table 14), so the constraint is silent on most of the genome and must be a term that switches off outside the model rather than a hard requirement. And betaxanthin is already the strand closest to its ceiling, so the headroom the constraint is competing for is the smallest of any strand: 0.430 against 0.914. Expect it to earn its place by making the mechanism legible and by extending to targets with no screen at all, rather than by moving betaxanthin's number a lot.

Not an expression rescue. The same layer is attractive for expression on the argument that metabolism constrains transcription. table 7 is the reason to be cautious: expression's binding constraint is 1,484 perturbations each seen once, and a metabolic prior does not add perturbations. Gate it on the same evidence standard the amino-acid coupling met, a measurable relationship in the data first.

Reifying the gene node into mRNA and protein. Not yet. The measured case for it is thin: per-gene mRNA-to-protein correlation across strains has a median of 0.08, a ridge map between the two recovers 2 to 3.5 % of held-out variance, and the two datasets were measured in different media, so the decoupling that would justify separate nodes is partly a condition artifact. Splitting the node multiplies parameters on an axis the data does not yet resolve. The case changes when a same-media proteome and expression pair exists, or when perturbations become fine-grained enough that the central-dogma collapse is what breaks.

Masked-label imputation as its own deliverable. The masked-conditioning oracle reaches cross-study 0.4838 at $m = 1000$ against the model's 0.241, and 97.5 to 100.6 % of that gain survives removing a genotype predictor. It is a real capability and a defensible product surface, and it is not a genotype-to-expression result. Deciding whether to ship it as a separate thing is a decision, not an experiment.

8.4 What to do about Figure 3 and Figure 6 ✗

Figure 3. The manuscript currently claims $r = 0.543$ for expression and 0.619 for morphology inside a paragraph marked `[FILLER - R3.]`. The measured values are 0.241 and 0.082. Neither placeholder should survive into a draft anyone reads. The honest version of the panel is the ceiling-relative one: expression at 31 % of a measured 0.775, morphology at 13 % of a measured 0.611 *on a quarter of its data*, with the fitness panel carrying the strong result. **A** 0.24 **per-gene Pearson is not a Figure 3 headline.**

Two of the three cheapest routes to a better one are not modeling work at all. Morphology has never seen more than 1,161 of its 4,718 strains in any of 397 runs, so the full-scale run is a query change and is the largest single change available to the figure. And the graph-prior probe decides whether nine attention heads are enforcing a premise nobody has checked. Both land before any arm ladder is worth launching; section 9 sizes what follows.

Figure 6. The betaxanthin case is the strongest thing in this document and it does not depend on beating anyone on accuracy. The argument is coverage: yeast-GEM reaches 19 % of the screen and misses 73 % of its high producers, and CGT finds high producers in the part a flux method cannot represent at 4.4× over base rate.

The last panel is where the figure either becomes an argument or stays a list, and getting its *structure* right is the open work. The sequence that carries it: setup diagram; betaxanthin performance against its 0.914 ceiling; the coverage gap; the capability gap; the Flux Cone Learning comparison, framed as a tie on the genes both can score; and then the structural panel. That last one is the constraint-based layer above: the amino-acid coupling is real in the data and unused by a plain auxiliary head, and a stoichiometric constraint over yeast-GEM is what would close it. Shown that way the amino-acid material is the *motivation* for the mechanism rather than a failed joint-training result, which is the only framing in which a negative belongs in a figure.

Keep the option text in the draw.io boxes. Several of those panels name alternatives that have not been run, and a composition that still reads when one of them does not arrive is worth more than a tidier one that does not.

9 The expression and morphology campaign ✗

The next campaign is expression plus morphology, for Figure 3. What follows is sized in GPU-days rather than in arms, because the measured cost per run is what makes most possible designs unaffordable.

9.1 The budget arithmetic, measured ✗

run	epochs	s/epoch	wall clock	peak at
v9 `M_fine` (masked)	9,999	32.9	**91.4 h = 3.81 d**	epoch 9,674 (97 %)
v8 `V_basis64`	4,106	42.0	47.9 h = 2.00 d	epoch 3,921 (95 %)

Table 22: One expression run is a multi-day object. Both runs peaked in their last few percent, so neither bracketed its own maximum.

At ≈ 35 s/epoch, one GPU delivers ≈ 2,500 epochs per day. So with G GPUs for D days the campaign has

$$\text{arm-slots} = \frac{G \cdot D \cdot 2{,}500}{E}$$

runs, where E is the per-arm epoch budget. **Every design decision below is a consequence of that fraction**, so it should be evaluated with the real G and D before anything is launched.

9.2 Set the epoch budget at 4,000, and the reason is measurable ✗

The previous round's failure was scoring arms at 300 epochs against a curve that had not turned. The opposite failure is affordable only once: 10,000 epochs is 3.8 days per arm, which at four GPUs for four days buys four runs and no replicates.

$E = 4{,}000$ is the defensible middle, and not by taste. The v8 run **peaked at epoch 3,921** of 4,106, so a pair-rank arm has been observed to reach its maximum inside that budget. It costs ≈ 1.6 days per run, so four GPUs for four days buys ≈ 10 slots.

What $E = 4{,}000$ gives up, stated plainly. The v9 masked run was still climbing at 9,674 and reached 0.2362; truncating it at 4,000 would have scored it near 0.21. So a 4,000-epoch comparison is fair between arms and **understates** the absolute number by roughly 0.02 to 0.03. That is the right trade for an arm comparison and the wrong one for a headline figure, which is why the winning arm gets one long run afterward.

9.3 Seeds: two per arm, and the noise floor is already known ✗

Expression's across-initialization spread was measured at ≈ 0.006 ($n = 8$, fixed split) in wave 5, an order of magnitude below the betaxanthin strand's 0.030 (table 21), because the split is pinned so `seed` varies weights only. At $\sigma = 0.006$, two seeds give a paired standard error of ≈ 0.004, so a 0.015 effect is nearly 4σ. **Two seeds per arm is enough**; three would buy precision the mechanism differences do not need.

9.4 Phase A, the objective, and why it goes first ✗

section 2.3 found that the quantile loss reaches its minimum at epoch 463 and rises for the next 9,500 while Pearson climbs, and that the model never gets `nmse` below 1. A mechanism arm evaluated under an objective that is optimizing something else cannot separate "the mechanism does not help" from "the objective does not reward it", so the objective is settled first.

Report `nmse` and the SD ratio on every arm, not just Pearson. The calibration identity in section 2.3 means an arm can raise Pearson while leaving `nmse` above 1, and that is a different result from one that fixes both. Both numbers are already logged.

Free, and worth doing on day one. Apply the post-hoc rescale by r/s to the existing best checkpoint. It changes no correlation and moves `nmse` from 1.010 to 0.944. It is an evening's work and it removes the "loses to the mean" objection from the figure regardless of what the campaign finds.

arm	objective	what a win means
`O_ref`	quantile (current)	reference
`O_zmse`	MSE on per-gene z-scored targets	"predict the mean" scores 0 rather than winning; already implemented as `standardize_per_feature_target`
`O_corr`	correlation objective	optimizing the quantity actually scored

Table 23: Phase A. Three arms, two seeds, $E = 4{,}000$: six slots, ≈ 10 GPU-days.

9.5 Phase B, the pair term, under the winning objective ✗

The reference operator provably cannot express any dependence on the pair (deleted gene, reporter gene), and masked unmasking does not supply one at $k = 0$. The ladder is the mechanism axis.

arm	pair rank	note
`P_ref`	0	additive reference
`P_basis64`	64	the current best-per-compute: 0.2274 in 2.0 days
`P_film`	90	diagonal multiplicative, identity at init
`P_graph`	–	masked graph attention applied **after** the perturbation operator, so the pair term is a function of network distance. **Not built.**

Table 24: Phase B. Four arms, two seeds: eight slots, ≈ 13 GPU-days. `P_graph` requires new code and is gated on the graph-prior probe below.

Gate `P_graph` on the probe, and run the probe this week. The probe (section 7) costs minutes of CPU and asks whether deleting gene X actually perturbs X's neighbors on each of the nine graphs. If the answer is no, `P_graph` is building a mechanism on a false premise and the nine attention heads currently spent enforcing that premise should be freed instead. **Nothing in the campaign is more decision-relevant per CPU-minute**, and it has been specified and unbuilt since 2026.07.26.

9.6 Morphology runs alongside, and it is a different problem ✗

Morphology is not a smaller copy of expression. Its target is 278-dimensional rather than 6,127, no morphology run has ever passed 121 epochs, and every one of the 397 runs across eight projects trained on 1,161 strains (table 9) out of a 4,718-strain screen.

1. **Drop `require_modalities` for the morphology-only arm** and rebuild the dataset on all 4,718 Ohya deletions. This is a query and config change, and it is the only way the 0.0824 becomes a statement about morphology rather than about a quarter of it. **Do this first**; it is the largest single change available to the figure.
2. **Measure the per-epoch cost on the first run.** It is not known: the only figure on record is a tiny-model smoke at ≈ 47 s/epoch against expression's ≈ 15 in the same smoke, which would make morphology the more expensive of the two despite the smaller target. Until it is measured, the slot arithmetic above does not apply to morphology.
3. **Run the scalar warm-up in parallel**, on `A113` (actin_n_ratio, ceiling 0.873, robust CV 2.37) or `C124_C`. It calibrates the epoch budget for the 278-feature target for a fraction of the cost.

9.7 The joint question, which is the one Figure 3 is about ✗

Does expression help morphology. The control already exists and is correct: `delta_joint_expr_morph_000.yaml` pins all three conditions to the 1,440 genotypes carrying both phenotypes, so `expr_morph` minus `morph` isolates the auxiliary head from a data-quantity difference. It has never been run past 87 epochs.

Run it at two scales, and do not conflate them. On the 1,440 shared strains it is the mechanism test it was designed to be. Paired with **fitness** instead of expression it keeps 4,220 of the 4,718 strains, three times the instance count, which is where the statistical power is. The two answer different questions and both are worth a slot; the fitness pairing is the one likely to produce a detectable effect.

The prior from the metabolite strands is not encouraging, and should be said out loud. The one auxiliary-head experiment run to completion anywhere in this project is the metabolome head on betaxanthin, and it reads -0.0265 ± 0.0159 (table 16). A second phenotype sharing an encoder has not yet been shown to help any first phenotype here. The expression-morphology pairing has a better prior than that one, because the two phenotypes share strains rather than merely genes, but the campaign should be designed to **measure** the effect rather than to confirm it, which means the control arm matters more than the joint arm.

9.8 What the campaign cannot fix ✗

Expression sits at 0.241 against a ceiling of 0.775, and the perturbation axis has 1,484 points with each gene deleted exactly once (table 7). No objective and no pair term changes that. If Phase A and Phase B both come back inside the noise floor, the honest reading is that the binding constraint is the data rather than the model, and the figure's expression panel should be built around the ceiling-relative statement and the imputation capability rather than around a genotype-to-expression number.